

USB-C PD Bench Power Supply

schematic evolution, v0.3 through v1.3

How the design moved

Rev	Dates	MCU	Sheet	What changed, and why
v0.3	24 Mar	CH32V003F4P6	single A3	First arrangement that worked on the bench. Four-switch buck–boost driven straight from MCU PWM, and a TIP120 NPN Darlington emitter follower as the pass device — the obvious way to build the stage. Set over UART: there is no panel, because every usable pin is already committed.
v0.4	25 Mar	CH32V003F4P6	single A3	The pass device turns round. The NPN follower needs its base two V_{BE} above the output, and the op-amp driving it runs from the same pre-regulator rail, so it saturates before the top of the range. A TIP127 PNP Darlington <i>sinks</i> base current instead and needs no headroom above VOUT at all.
v0.5	26 Mar	CH32V203C8T6	single A3	User interface added. The ’003 has 18 usable GPIO and 16 were spent on the power path, leaving two — not enough for an encoder, three keys, three lamps and a display. Moving up the same RISC-V family to the 48-pin part brings 37 GPIO and keeps the toolchain unchanged.
v0.6	27 Mar	CH32V203C8T6	single A3	Hardware over-current trip added after bench testing showed firmware alone cannot react to a short in time. This is the last thing that fits on one sheet.
10 days idle — prototype left on the bench				
v1.1	5–15 Apr	STM32G030C8T6	3-sheet	Redrawn as a hierarchy , and the MCU changes vendor. The +12 V gate rail becomes a real on-sheet boost instead of an off-sheet module, so the board can actually be built; RC snubbers are added across both switch nodes; the drivers gain proper bootstrap networks.
v1.2	25 Apr	STM32G030C8T6	3-sheet	0–36 V range, and the analogue nets move onto pins that have an ADC. v1.1 had put ADC_VBUS, ADC_VPRE, ADC_VOUT and ISNS.OUT on PD2, PD3, PC6 and PC7 — none of which has an ADC channel on this part, so not one of the four measurements could have been taken.
v1.3	25 May	STM32G030C8T6	3-sheet	The three faults that would have stopped the board working. See below.

What v1.3 fixes

Fault	Effect if built	Fix
Feedback loop never closed	The output divider tap and the error amplifier’s invert-ing input were two separate nets. U8A had no DC feed-back path at all: a bare integrator whose output ramps to a rail. The pass device would sit hard on or hard off — the supply would not regulate, it would latch.	The divider tap and the amplifier input carry one name, so they are one node. $V_{OUT} = 16 \times V_{REF}$, which is what the firmware already assumes.
High-side driver returns open	Each IRS2104’s VS pin sat on a net of its own rather than on the switch node it references. The boot-strap capacitor cannot refresh, so neither high-side FET switches.	VS ties to the power switch node itself.
Current sense saturated at 1.32 A	$0.05\,\Omega \times 50 = 2.5\,\text{V/A}$ reaches the 3.3 V ADC ceiling at 1.32 A against a 3 A rating. Above that the reading sticks, so the current limit, the CC indicator and the adaptive headroom all stop responding while the ana-logue loop keeps delivering current.	$0.01\,\Omega$ with an INA181A3 gives 1 V/A: full scale 3.3 A, and a fifth of the shunt dissipation. The trip divider becomes 22k/39k, putting the 3.2 A threshold at 3.2 V — reachable from the 5 V rail, which it was not before.

1 – USB-C PD INPUT

The diagram illustrates the USB-C PD input circuit. A USB-C connector (J1) is connected to a shield (S1) and ground (A12). The VBUS pin (A9) is connected to a 3 A hold fuse (F1). The CC1 (A5) and CC2 (B5) pins are connected to the CC1 and CC2 pins of the CH224K IC (U1). The CH224K IC is connected to the VBUS, VDD, and PG pins. The VDD pin is connected to a 100uF/35V capacitor (C1). The PG pin is connected to a 100k 1% resistor (R1) and a 10k 1% resistor (R2). The DM (X5) and DP (X4) pins are connected to the DM and DP pins of the CH224K IC. The CH224K IC is also connected to the PD_PG, CFG1, CFG2, and CFG3 pins.

CH224K requests 5 / 9 / 12 / 15 / 20 V through CFG1...3. The MCU re-reads ADC_VBUS 500 ms after plug-in; no rise means a legacy 5 V port.

2 – 4-SWITCH BUCK-BOOST (MCU PWM – no switcher IC)

3 – LINEAR POST-REG + CURRENT SENSE

The circuit diagram illustrates a linear post-regulator with current sensing. It consists of the following components and connections:

- Op-Amp U4A (LM358):** configured as a voltage follower. Its non-inverting input (+) is connected to **VREF**. Its inverting input (-) is connected to the feedback node **FB**. Its output (pin 1) drives the base of the NPN Darlington emitter follower **Q5 (TIP120)**. The emitter of **Q5** is connected to ground.
- Op-Amp U4B (LM358):** configured as a voltage follower. Its non-inverting input (+) is connected to **VOUT**. Its inverting input (-) is connected to ground. Its output (pin 7) provides the **ADC_ISNS** signal.
- Current Sense:** The collector of **Q5** is connected to the load through resistor **R3 (0R05 2W)**. A sense resistor **R5 (39k 1%)** is connected between the collector of **Q5** and the **FB** node. Resistor **R6 (10k 1%)** is connected between the **FB** node and ground.
- Output Filter:** A capacitor **C5 (220u/25V)** is connected between the output node (after **R3**) and ground.
- Input and Biasing:** The input **VPRE** is connected to the base of **Q5** through resistor **R7 (100k 1%)**. Resistor **R8 (10k 1%)** is connected between the base of **Q5** and ground. The **Q5_BASE** signal is connected to the base of **Q5** through resistor **R4 (1k)**.
- Output:** The output voltage **VOUT** is taken from the output node after the filter capacitor **C5**.

Legend:

- VPRE: Input voltage
- VREF: Reference voltage
- VOUT: Output voltage
- ADC_ISNS: Current sense output
- Q5_BASE: Base of Q5
- FB: Feedback node
- Q5: NPN Darlington emitter follower (TIP120)
- R3: 0R05 2W
- R4: 1k
- R5: 39k 1%
- R6: 10k 1%
- R7: 100k 1%
- R8: 10k 1%
- C5: 220u/25V
- U4A: LM358
- U4B: LM358

Notes:

- NPN Darlington emitter follower – the op-amp drives the base, the emitter follows. Shunt in the output return, second LM358 half lifts it into ADC range.

4 – +5 V LDO

U5
LP2950–5.0

The diagram shows the connection of the LP2950-5.0 LDO. The input pin (3, VI) is connected to the VBUS supply. A bypass capacitor C6 (10uF) is connected between the input pin and ground. The output pin (1, VO) is connected to the +5V output. A bypass capacitor C7 (10uF) is connected between the output pin and ground. The ground pin (2, GND) is connected to the common ground. A red arrow indicates the output voltage is +5.0V.

5 – MCU

U6
CH32V003F4P6

Pinout details:

- Pin 9: VDD (connected to C8 100nF)
- Pin 7: VSS
- Pin 10: CFG1
- Pin 11: CFG2
- Pin 12: CFG3
- Pin 13: FB
- Pin 14: PWM_REF
- Pin 15: EN_A
- Pin 16: EN_B
- Pin 17: PC0
- Pin 18: PC1
- Pin 19: PC2
- Pin 20: PC3
- Pin 1: PD0
- Pin 2: PD1
- Pin 3: PD2
- Pin 4: PD3
- Pin 5: PD4
- Pin 6: PD5
- Pin 8: PD6
- Pin 18: PD7
- Pin 19: PA1
- Pin 20: PA2
- Pin 8: PD_PG
- Pin 18: SWIO
- Pin 19: PWM_A
- Pin 20: PWM_B
- Pin 1: ADC_ISNS
- Pin 2: UART_TX
- Pin 3: UART_RX
- Pin 5: ADC_VPRE
- Pin 6: ADC_VBUS

6 – PROG / UART

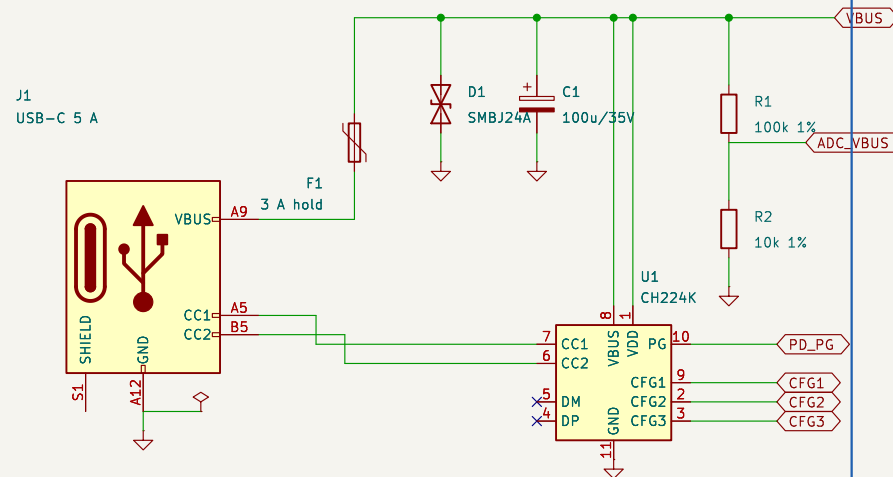
J3
PROG

The diagram shows a 4-pin connector labeled J3. The pins are numbered 1 through 4. Pin 1 is connected to SWIO. Pin 2 is connected to UART_TX. Pin 3 is connected to UART_RX. Pin 4 is connected to Ground (indicated by a triangle symbol).

No switcher IC – the MCU drives all four gates. BUCK: leg A chops, Q4 parked on. BOOST: Q1 parked on, leg B chops. Dead-time and bootstrap refresh in firmware.

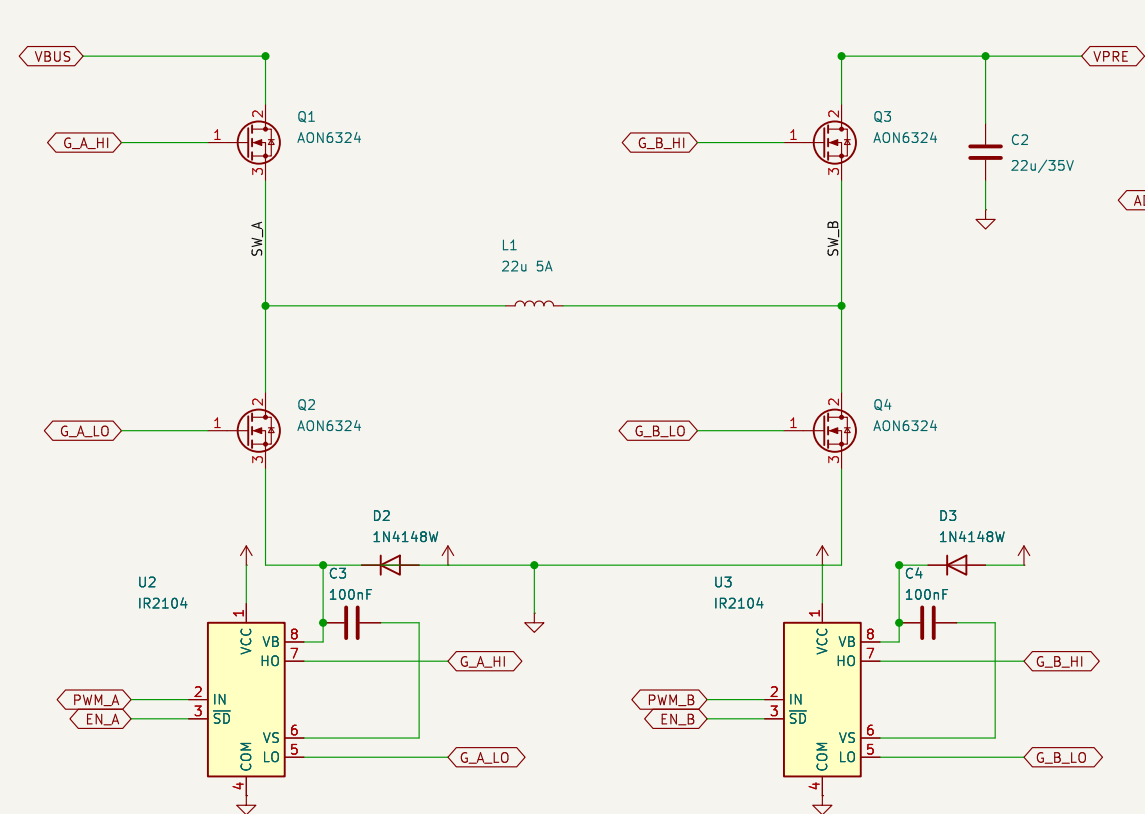
Title: USB-C PD Bench PSU - 4-Switch Buck-Boost + PNP Linear Post-Reg

1 - USB-C PD INPUT

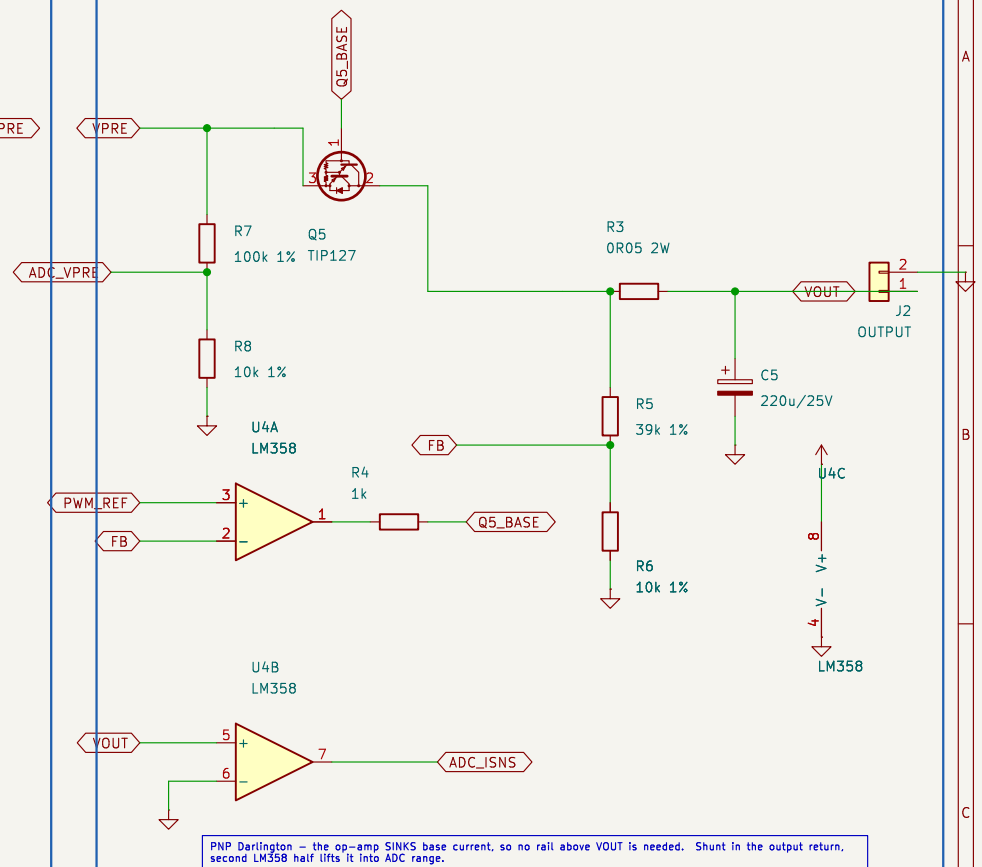


CH224K requests 5 / 9 / 12 / 15 / 20 V through CFG1..3. The MCU re-reads ADC_VBUS 500 ms after plug-in; no rise means a legacy 5 V port..

2 - 4-SWITCH BUCK-BOOST (MCU PWM - no switcher IC)

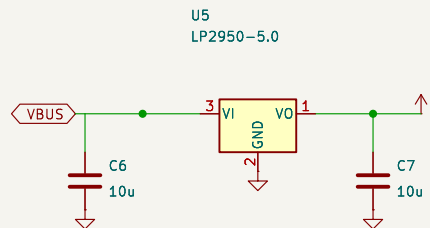


3 - LINEAR POST-REG + CURRENT SENSE

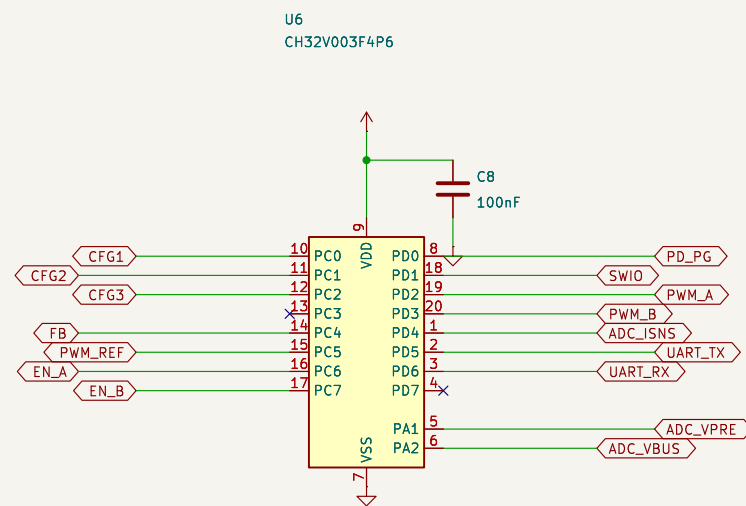


PNP Darlington - the op-amp SINKS base current, so no rail above VOUT is needed. Shunt in the output return, second LM358 half lifts it into ADC range.

4 - +5 V LDO

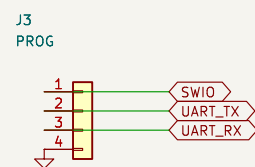


5 - MCU



No switcher IC - the MCU drives all four gates. BUCK: leg A chops, Q4 parked on. BOOST: Q1 parked on, leg B chops. Dead-time and bootstrap refresh in firmware.

6 - PROG / UART



[illegible]

2 – 4-SWITCH BUCK-BOOST (MCU PWM – no switcher IC)

3 - LINEAR POST-REG + CURRENT SENSE

The circuit diagram illustrates a linear post-regulator with current sensing capability. It features two LM358 op-amp halves (U4A and U4B) configured as voltage followers.

- Op-Amp U4A:** The non-inverting input (+) is connected to the feedback signal (FB). The inverting input (-) is connected to the output (VOUT) through a resistor R4 (1k). The output of U4A drives the base of a PNP Darlington transistor (Q5, TIP127).
- Op-Amp U4B:** The non-inverting input (+) is connected to the output (VOUT). The inverting input (-) is connected to ground. Its output provides the current sense signal (ADC_ISNS).
- Transistor Q5:** A PNP Darlington transistor used for current regulation. Its emitter is connected to the output (VOUT), and its collector is connected to the load through a resistor R3 (0R05 2W).
- Current Sensing:** A resistor R6 (10k 1%) is placed in series with the load return path. The voltage drop across R6 is sensed by U4B.
- Feedback Network:** Resistor R5 (39k 1%) connects the output (VOUT) back to the FB input of U4A. Resistor R7 (100k 1%) connects VPRE to the FB input, and resistor R8 (10k 1%) connects VREF to the FB input.
- Output Filter:** A capacitor C5 (220µ/25V) is connected from the output (VOUT) to ground.

PNP Darlington – the op-amp SINKS base current, so no rail above VOUT is needed. Shunt in the output return, second LM358 half lifts it into ADC range.

4 – +5 V LDO

U5
LP2950-5.0

The diagram shows the LP2950-5.0 LDO regulator (U5) connected between VBUS and the output line. The input (VI, pin 3) is connected to VBUS and has a bypass capacitor C6 (10uF). The ground (GND, pin 2) is connected to ground. The output (VO, pin 1) is connected to the output line and has a bypass capacitor C7 (10uF).

7 – USER INTERFACE

J4
OLED 0.96in I2C

1
2
3
4

I2C_SCL
I2C_SDA

CU drives all four gates. BUCK: leg A chops. Q4 parked on. B005T: Q1 parked on, leg B chops. refresh in firmware.

SW1
EC11

ENC_A
ENC_B

A
C
B

S1
S2

ENC_SW

SW2
OUTPUT

SW3
SELECT

SW4
BACK

BTN_OUT
BTN_SEL
BTN_BACK

D4
red

D5
amber

D6
green

R11
1k

R12
1k

R13
1k

LED_OUT
LED_EC
LED_HB

6 – PROG / UART

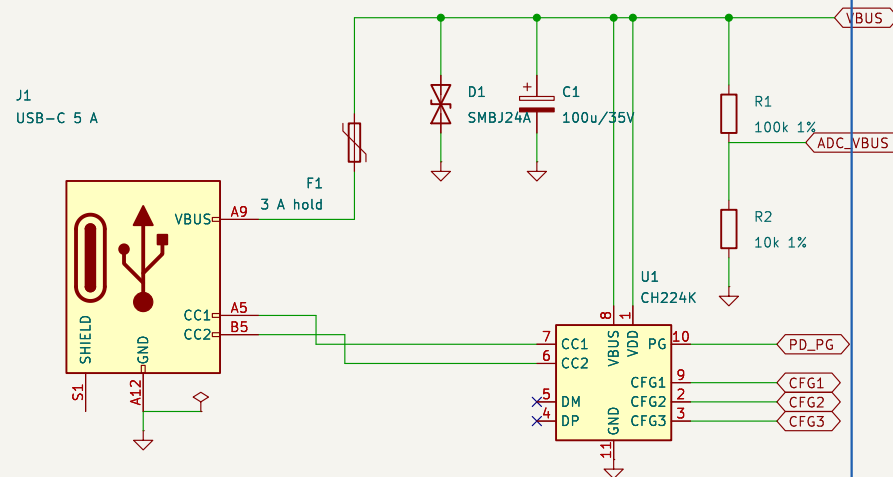
J3
PROG



The diagram shows a 4-pin connector labeled J3. The pins are numbered 1, 2, 3, and 4. Pin 1 is connected to SWDIO, Pin 2 is connected to SWCLK, and Pin 3 is connected to UART_TX. Pin 4 is connected to ground.

Post-1945

1 - USB-C PD INPUT

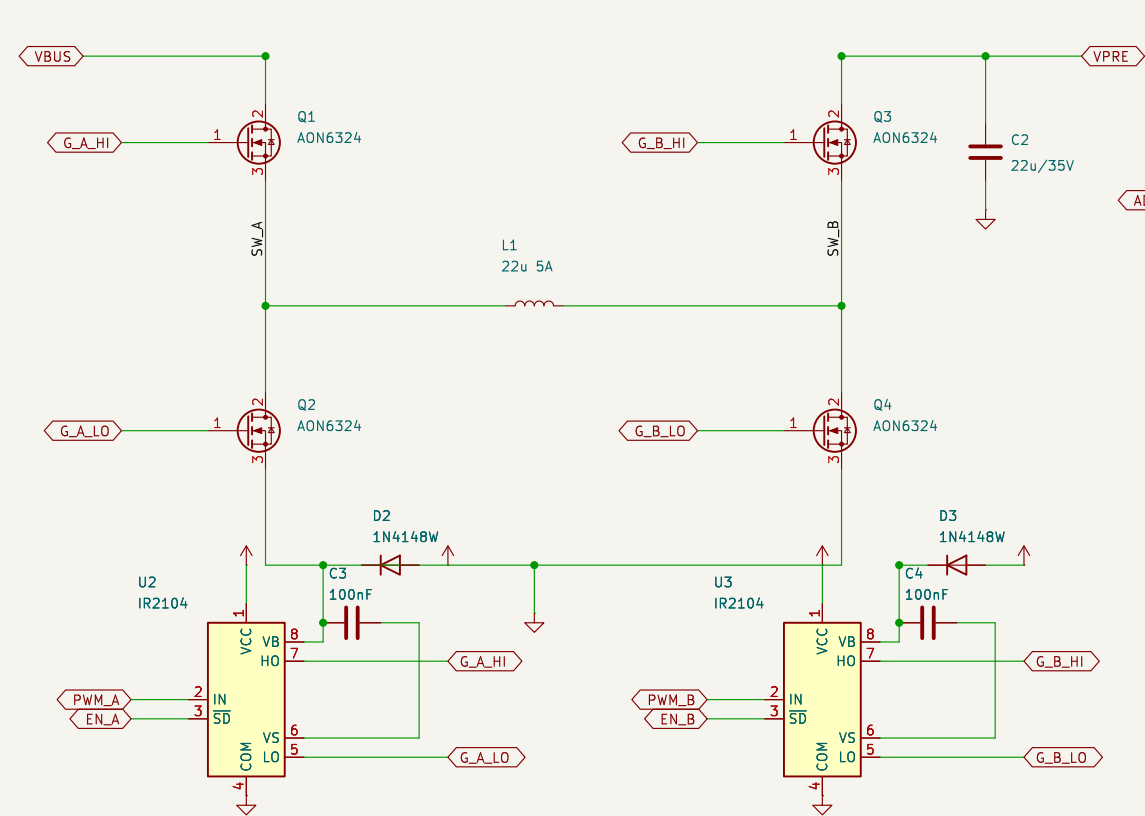


CH224K requests 5 / 9 / 12 / 15 / 20 V through CFG1..3. The MCU re-reads ADC_VBUS 500 ms after plug-in; no rise means a legacy 5 V port.

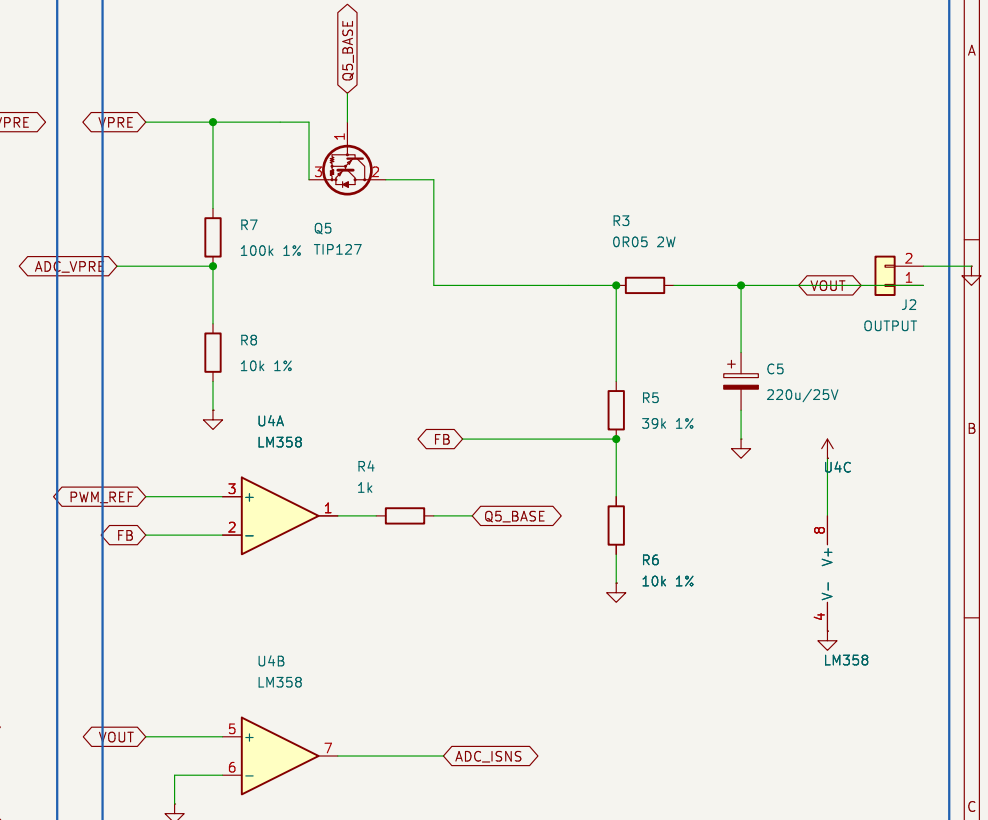
The panel forced the MCU change; the '003 had two GPIO left, and the panel needs eleven. The CH32V203C0T6 keeps the same core and toolchain, and brings 37.

Firmware alone cannot react fast enough to a short, so this comparator pulls the driver enable low directly - no code in the path.

2 - 4-SWITCH BUCK-BOOST (MCU PWM - no switcher IC)

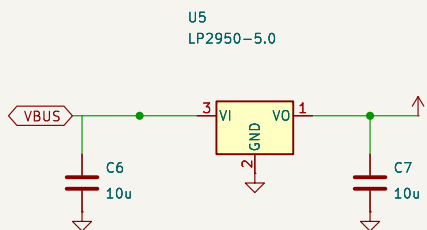


3 - LINEAR POST-REG + CURRENT SENSE

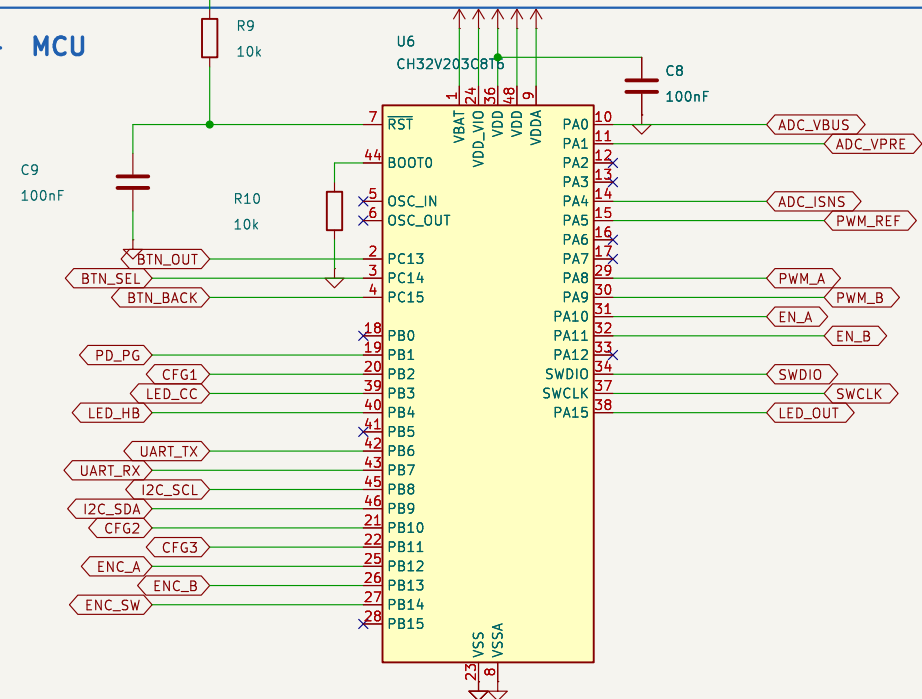


PNP Darlington - the op-amp SINKS base current, so no rail above VOUT is needed. Shunt in the output return, second LM358 half lifts it into ADC range.

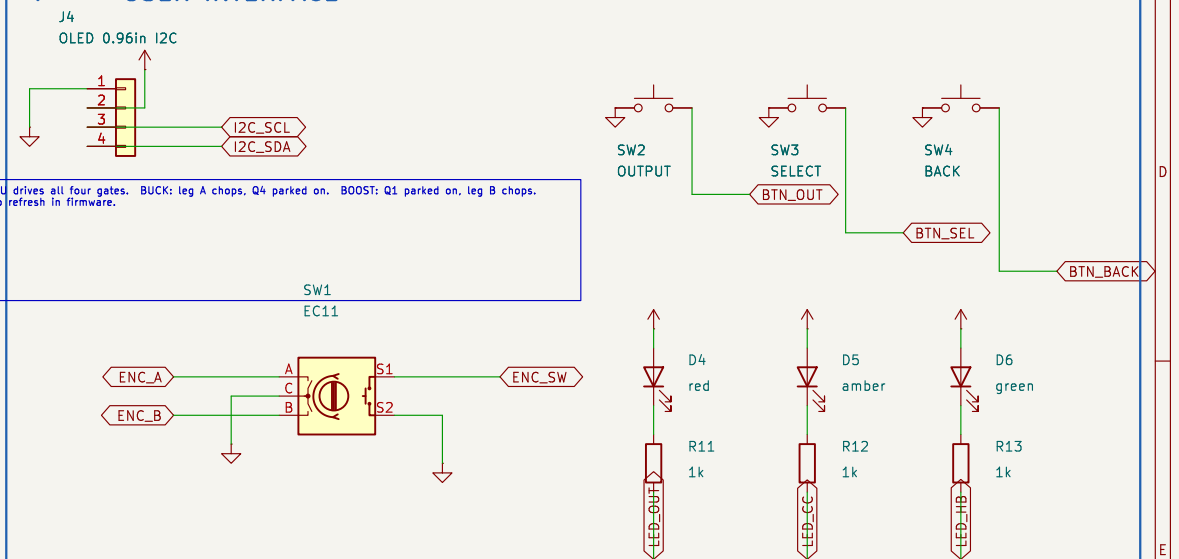
4 - +5 V LDO



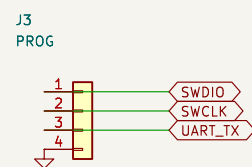
5 - MCU



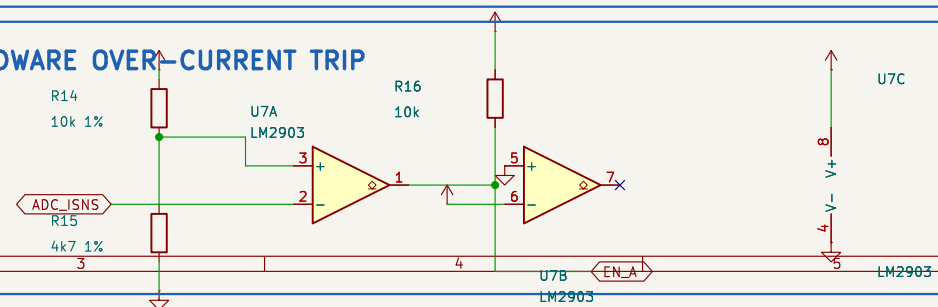
7 - USER INTERFACE



6 - PROG / UART



8 - HARDWARE OVER-CURRENT TRIP



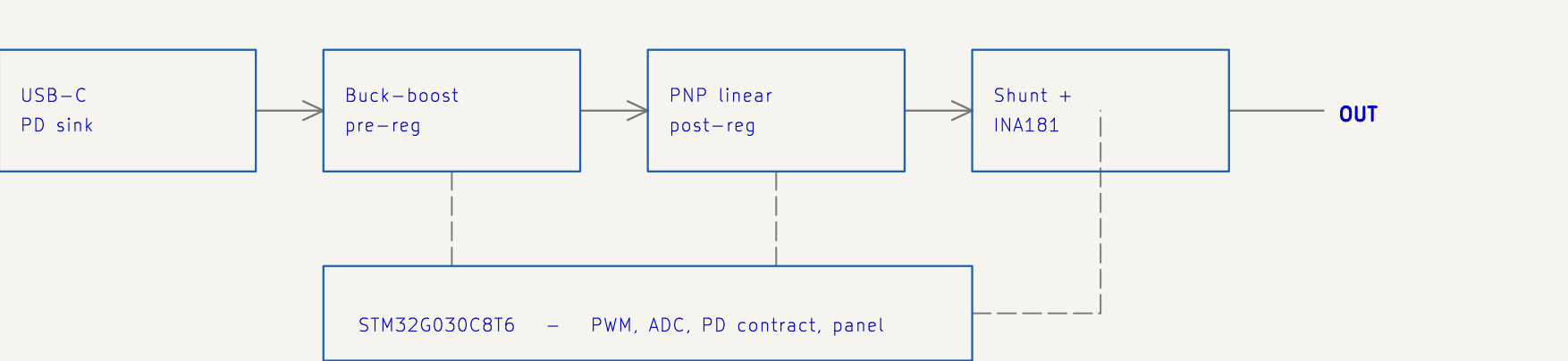
USB-C PD BENCH POWER SUPPLY

0 – 36 V / 0 – 3 A adjustable bench supply from a USB-C PD source

SPECIFICATION

Input	USB-C PD sink, 5 / 9 / 12 / 15 / 20 V contracts
Pre-regulator	4-switch synchronous buck-boost, 250 kHz
Post-regulator	PNP Darlington linear stage, tracking headroom
Output	0 – 36 V, 0 – 3 A
Set resolution	10 mV / 10 mA, PWM reference + 2-pole RC
Readback	voltage and current, 12-bit ADC
Protection	hardware over-current trip, latching
Modes	constant voltage, constant current, battery emulation
Control	STM32G030C8T6, OLED + rotary encoder + 3 keys

BLOCK DIAGRAM



REVISION HISTORY

Rev	MCU	Change
v0.3	CH32V003F4P6	first bench build; NPN Darlington follower
v0.4	CH32V003F4P6	pass device turned round to a PNP
v0.5	CH32V203C8T6	front panel added; the '003 runs out of pins
v0.6	CH32V203C8T6	hardware over-current trip
v1.1	STM32G030C8T6	redrawn as a hierarchy; on-sheet 12 V rail

DESIGNED AND DRAWN BY

Mohammadreza Safaeian

Embedded Systems & Hardware Engineer

m.re.safaeian@gmail.com

github.com/mohammadrezasafaeian

(c) 2026 Mohammadreza Safaeian. All rights reserved.

Power Path

1	input / PD trigger
2	+12 V gate rail
3	buck-boost power stage
4	gate drivers
5	over-current trip

Control and UI

7	linear post-regulator
8	current sense
9	MCU
10	user interface
11	programming header

github.com/mohammadrezasafaeian | (c) 2026 Mohammadreza Safaeian. All rights reserved.

Designed and drawn by Mohammadreza Safaeian – m.re.safaeian@gmail.com

Title page – specification, block diagram, sheet index

Mohammadreza Safaeian – personal project

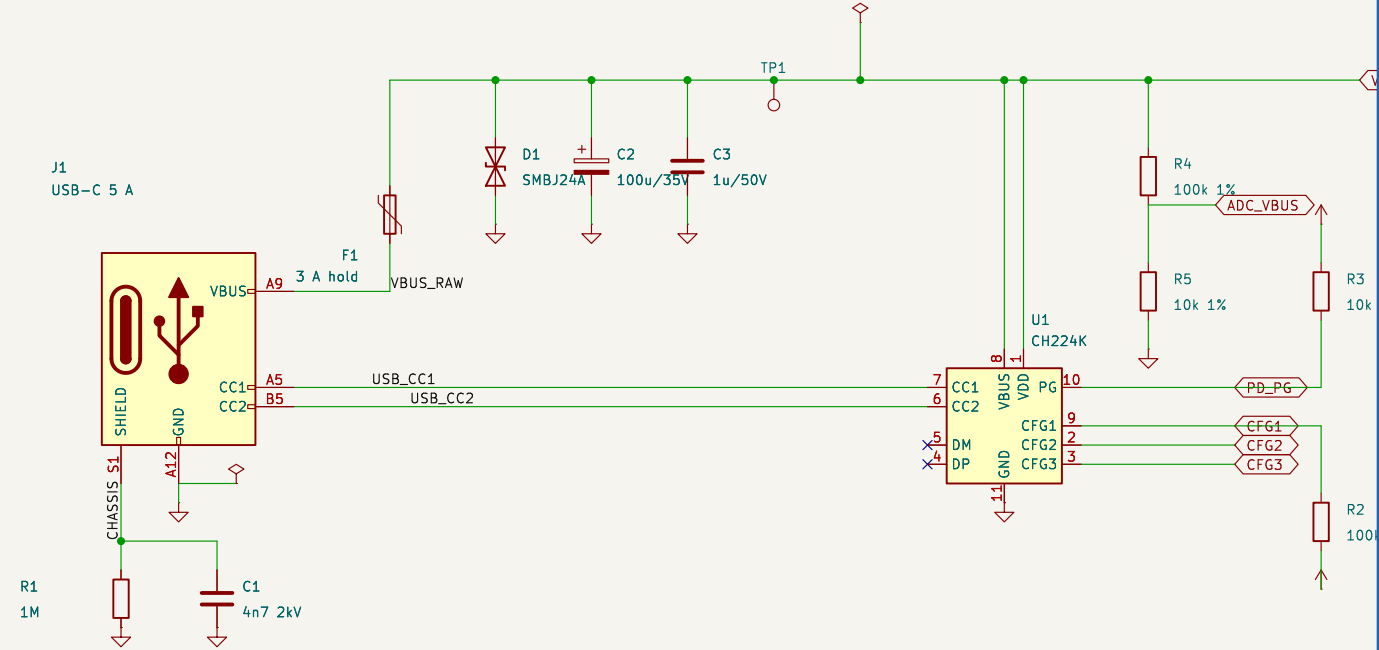
Sheet: /
File: usbc_pd_psu.kicad_sch

Title: USB-C PD Bench PSU – 4-Switch Buck-Boost + PNP Linear Post-reg

Size: A3 Date: 2026-04-15 Rev: v1.1

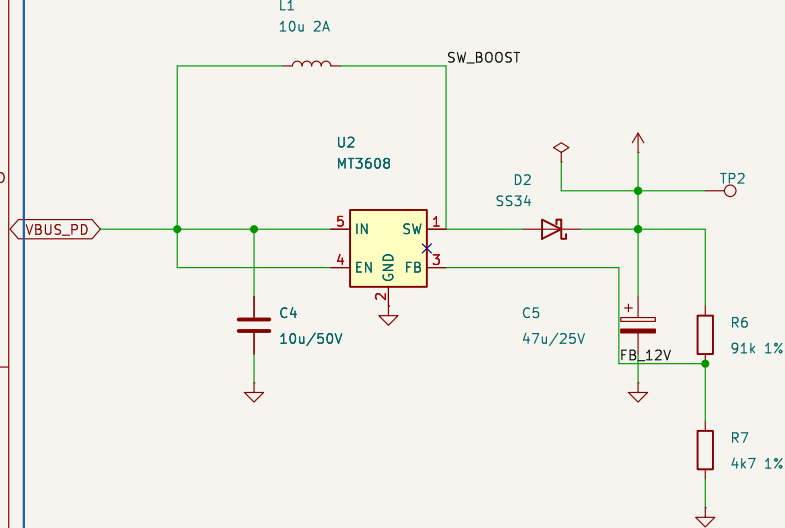
KiCad E.D.A. 9.0.2 Id: 1/3

1 – USB-C INPUT, PROTECTION & PD TRIGGER



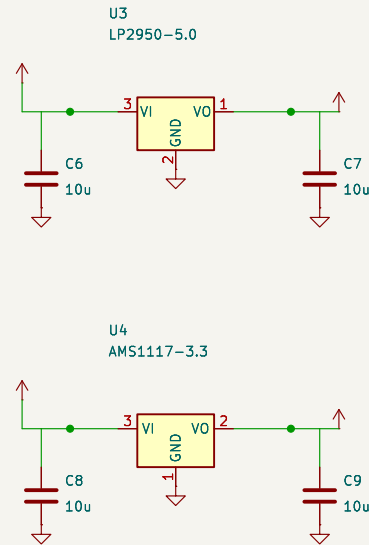
PD contract: CFG1/2/3 select 5 / 9 / 12 / 15 / 20 V. CFG1 is pulled high so a bare board requests 5 V before firmware runs. 500 ms after plug-in the MCU re-reads ADC_VBUS; no rise means a legacy 5 V port and the firmware clamps the output range instead of faulting. The PDO is re-selected as the setpoint changes (table on the title page), with the output disabled during renegotiation.

2 – +12 V GATE-DRIVE RAIL (MT3608 boost)



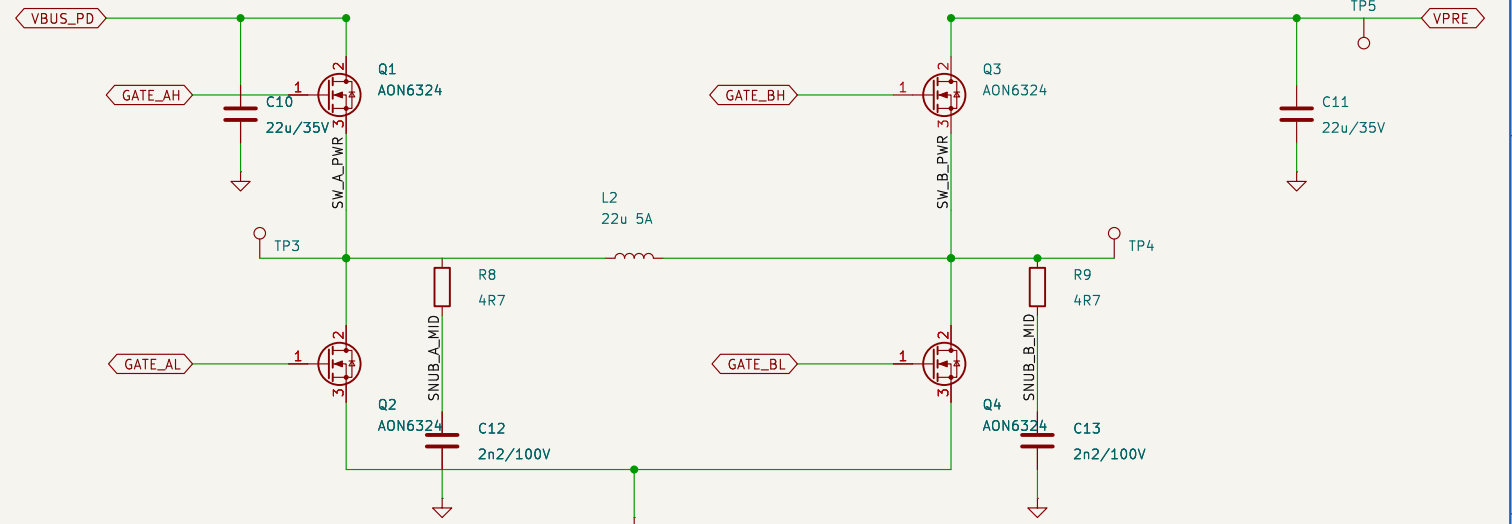
Why an on-board boost: the IRS2104 needs 10–12 V of gate drive (UVLO 8.9 V typ) to put the bridge FETs at their rated RDS(on). This 10 uH / SS34 stage makes 12 V at 150 mA from any contract between 5 V and 20 V, so the drivers behave the same on a legacy 5 V port as on a 20 V one.

6 – LOGIC SUPPLIES



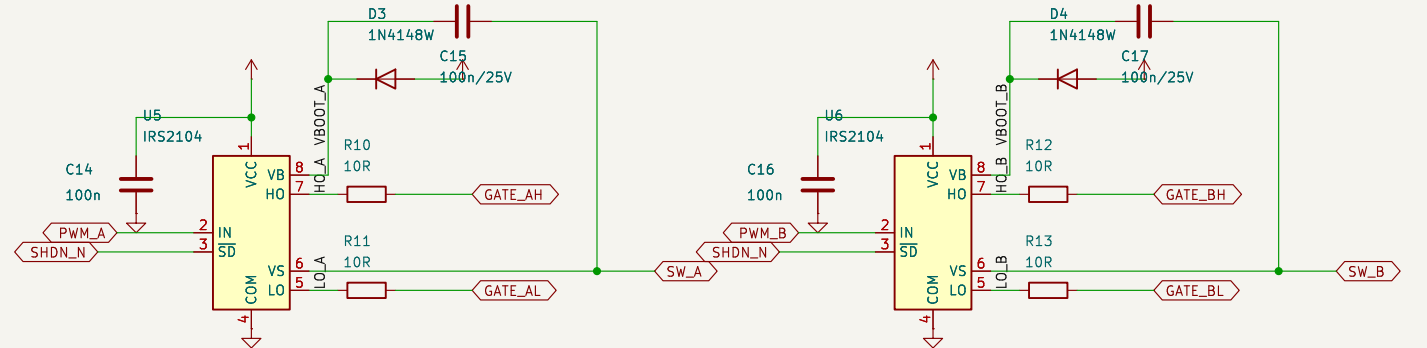
Fed from +12 V, not from VBUS: on a legacy port VBUS sits at ~4.75 V at the connector, below the LP2950 dropout for a 5 V output. Rails therefore come up identically on either port type.

3 – 4-SWITCH SYNCHRONOUS BUCK-BOOST POWER STAGE



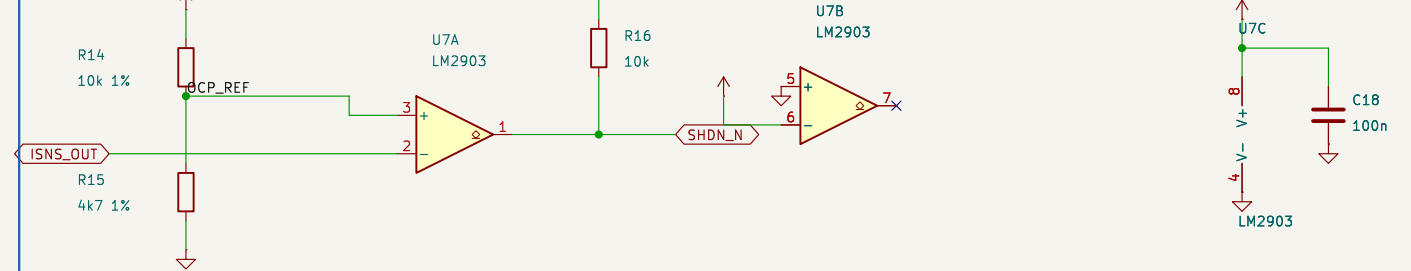
The 4R7 + 2n2 snubbers damp the switch-node ringing set by FET Coss and layout inductance. 60 V FETs, because a 20 V contract plus that ringing will exceed a 30 V part.

4 – HALF-BRIDGE GATE DRIVERS (IRS2104, 520 ns dead-time)



Each bootstrap capacitor returns to its own VS pin, which is tied to the switch node it references – that is what lets the cap refresh and the high-side FET turn on. 10R gate resistors slow the edges enough to keep the ringing in hand.

5 – HARDWARE OVER-CURRENT TRIP + LATCH (firmware-independent)



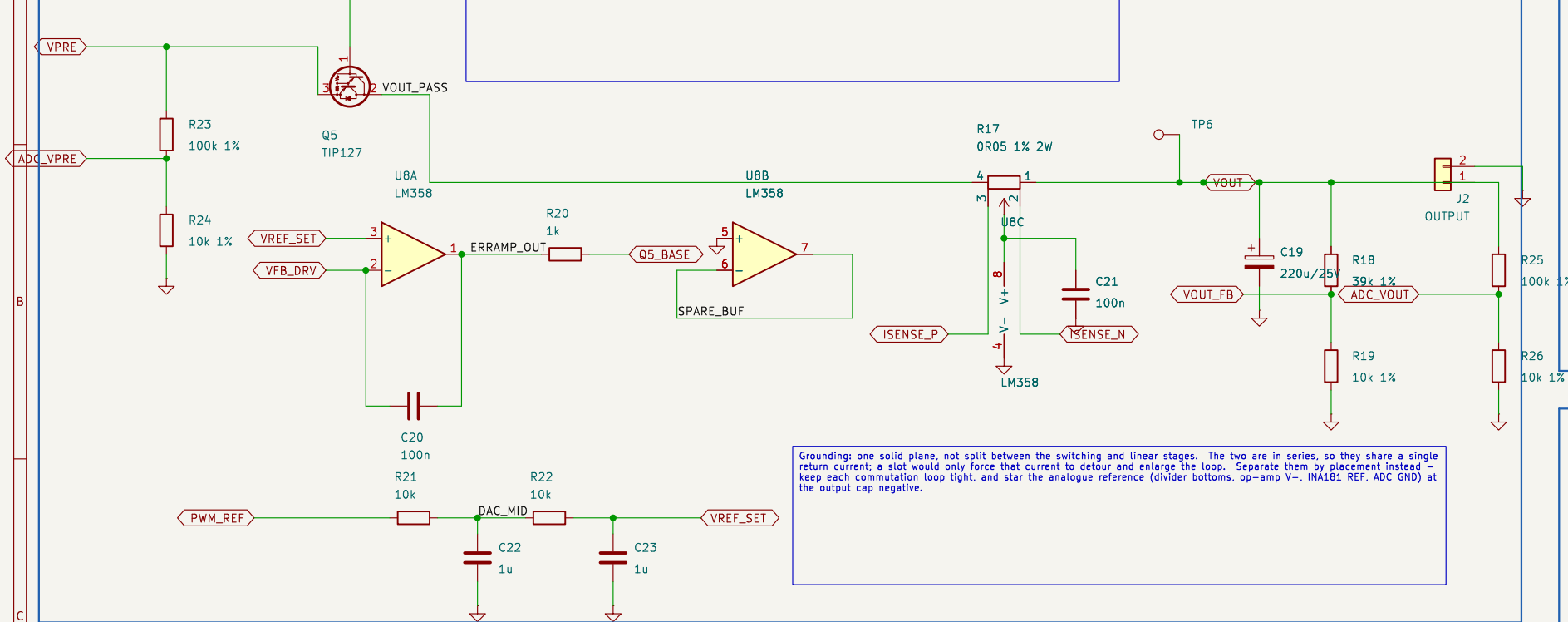
ISNS sits on the inverting input, so an over-current drives the open-collector output LOW and pulls SHDN_N down within ~1 us with no firmware involved. Both IRS2104 SD pins hang on SHDN_N, as does the MCU (open-drain), giving a wired-AND kill.

github.com/mohammadrezasafaeian | (c) 2026 Mohammadreza Safaeian. All rights reserved.
Designed and drawn by Mohammadreza Safaeian – m.re.safaeian@gmail.com
Sheet 1 of 2 – input, PD trigger, 12 V rail, buck-boost, drivers, OCP
Mohammadreza Safaeian – personal project

Sheet: /Power Path/
File: power.kicad_sch
Title: USB-C PD Bench PSU – 4-Switch Buck-Boost + PNP Linear Post-Regulator
Size: A3 Date: 2026-04-15 Rev: v1.1
KiCad E.D.A. 9.0.2 Id: 2/3

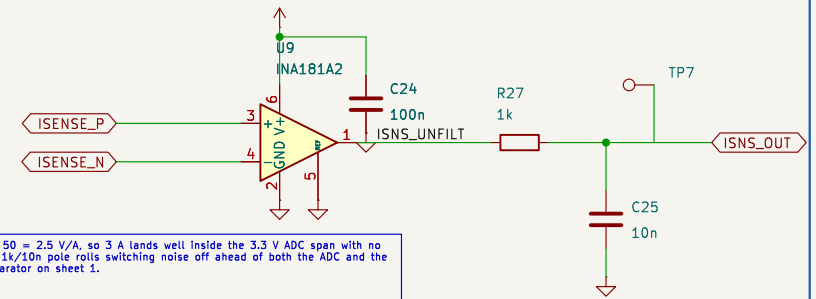
7 - PNP-DARLINGTON LINEAR POST-REGULATOR + OUTPUT

PNP Darlington pass device. The op-amp sinks base current, so the stage needs no headroom above VOUT — an NPN follower would need a base drive above the output and runs out at the top of the range. The shunt is high-side, between collector and VOUT, so the output negative stays at true ground and the INA181 reads correctly near 0 V.



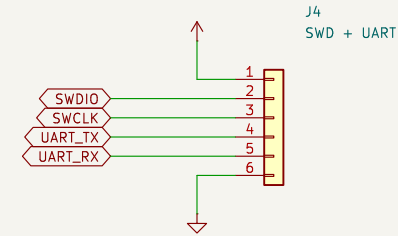
Grounding: one solid plane, not split between the switching and linear stages. The two are in series, so they share a single return current; a slot would only force that current to detour and enlarge the loop. Separate them by placement instead — keep each commutation loop tight, and star the analogue reference (divider bottoms, op-amp V-, INA181 REF, ADC GND) at the output cap negative.

8 - CURRENT-SENSE AMPLIFIER (INA181A2, gain 50)

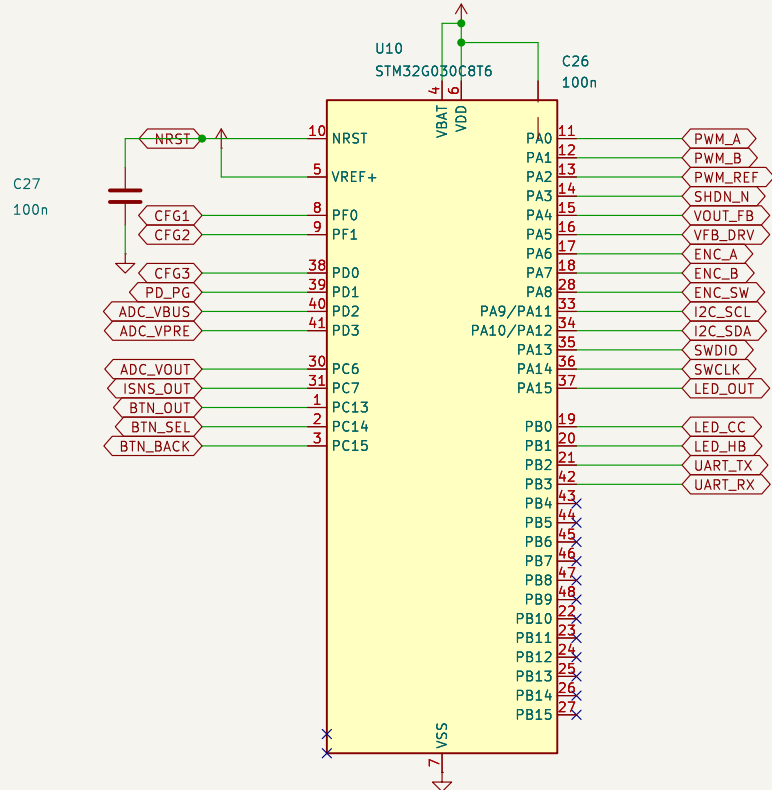


OR05 shunt $\times$ gain 50 = 2.5 V/A, so 3 A lands well inside the 3.3 V ADC span with no extra divider. The 1k/10n pole rolls switching noise off ahead of both the ADC and the over-current comparator on sheet 1.

11 - SWD + UART HEADER



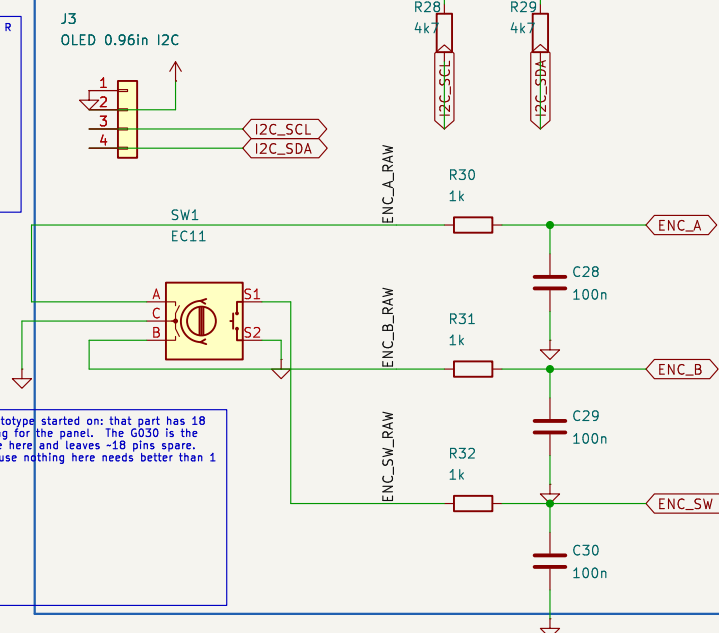
9 - MCU -- STM32G030C8T6 (LQFP-48)



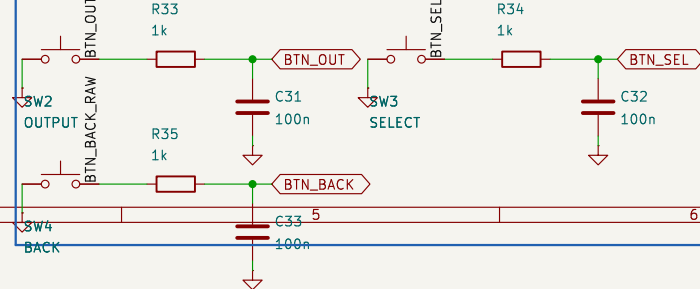
Indicator LEDs sink rather than source: +3V3 - LED - R - GPIO, so a pin left tri-stated at reset cannot back-feed the rail. PD_PG dark means the board fell back to a legacy 5 V port; SHDN_N lights on an over-current even with the MCU dead.

STM32G030C8T6, chosen over the CH32V003 the prototype started on: that part has 18 GPIO and 17 were already committed, leaving nothing for the panel. The G030 is the cheapest 48-pin Cortex-M0+ that is easy to source here and leaves 18 pins spare. Runs on the 16 MHz internal RC — no crystal, because nothing here needs better than 1 % timing. SWD on PA13/PA14.

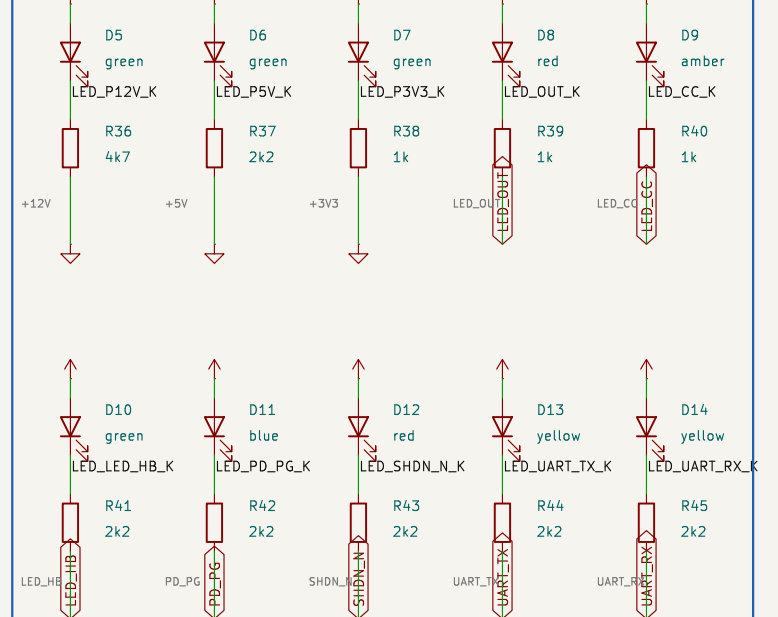
10 - UI: OLED + ROTARY ENCODER



13 - FRONT PANEL



12 - INDICATORS / LED TEST POINTS



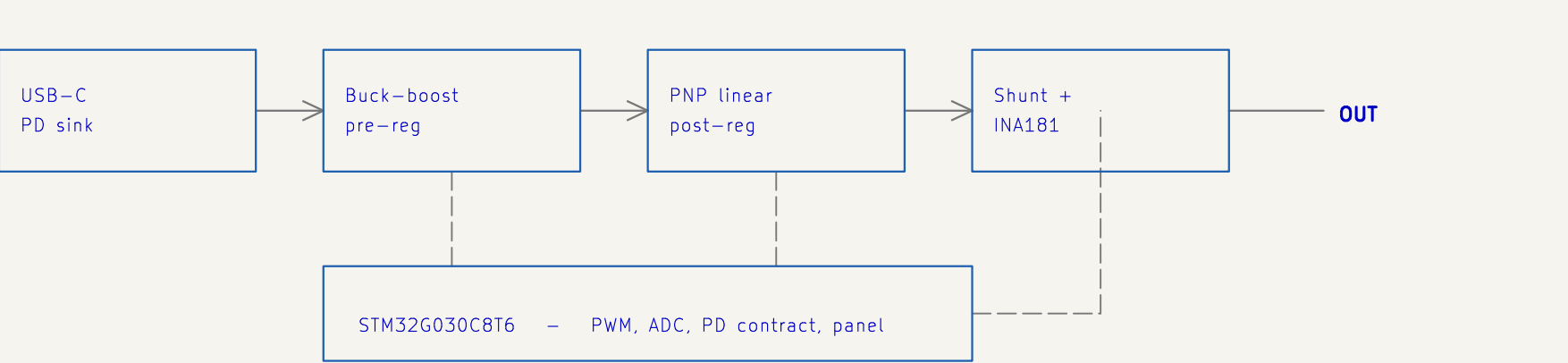
USB-C PD BENCH POWER SUPPLY

0 – 36 V / 0 – 3 A adjustable bench supply from a USB-C PD source

SPECIFICATION

Input	USB-C PD sink, 5 / 9 / 12 / 15 / 20 V contracts
Pre-regulator	4-switch synchronous buck-boost, 250 kHz
Post-regulator	PNP Darlington linear stage, tracking headroom
Output	0 – 36 V, 0 – 3 A
Set resolution	10 mV / 10 mA, PWM reference + 2-pole RC
Readback	voltage and current, 12-bit ADC
Protection	hardware over-current trip, latching
Modes	constant voltage, constant current, battery emulation
Control	STM32G030C8T6, OLED + rotary encoder + 3 keys

BLOCK DIAGRAM



REVISION HISTORY

Rev	MCU	Change
v0.3	CH32V003F4P6	first bench build; NPN Darlington follower
v0.4	CH32V003F4P6	pass device turned round to a PNP
v0.5	CH32V203C8T6	front panel added; the '003 runs out of pins
v0.6	CH32V203C8T6	hardware over-current trip
v1.1	STM32G030C8T6	redrawn as a hierarchy; on-sheet 12 V rail
v1.2	STM32G030C8T6	0–36 V range; analogue nets onto real ADC pins

DESIGNED AND DRAWN BY

Mohammadreza Safaeian

Embedded Systems & Hardware Engineer

m.re.safaeian@gmail.com

github.com/mohammadrezasafaeian

(c) 2026 Mohammadreza Safaeian. All rights reserved.

Power Path

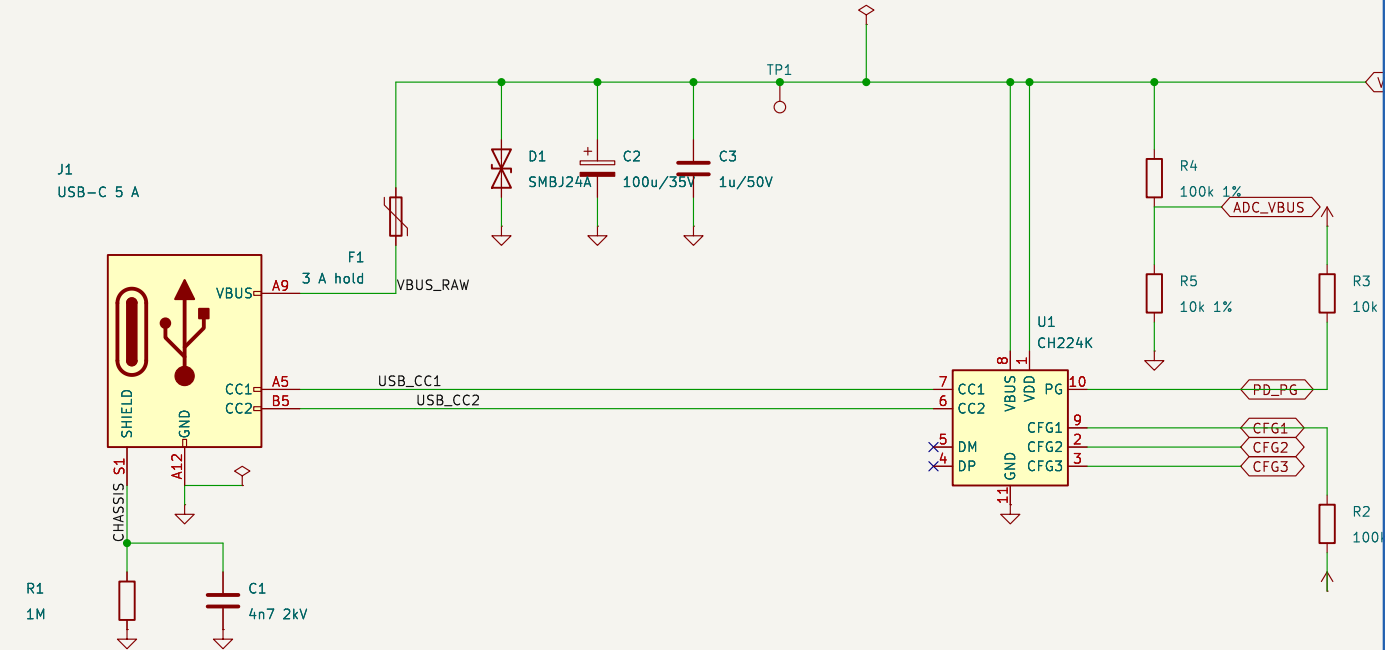
1	input / PD trigger
2	+12 V gate rail
3	buck-boost power stage
4	gate drivers
5	over-current trip

Control and UI

7	linear post-regulator
8	current sense
9	MCU
10	user interface
11	programming header

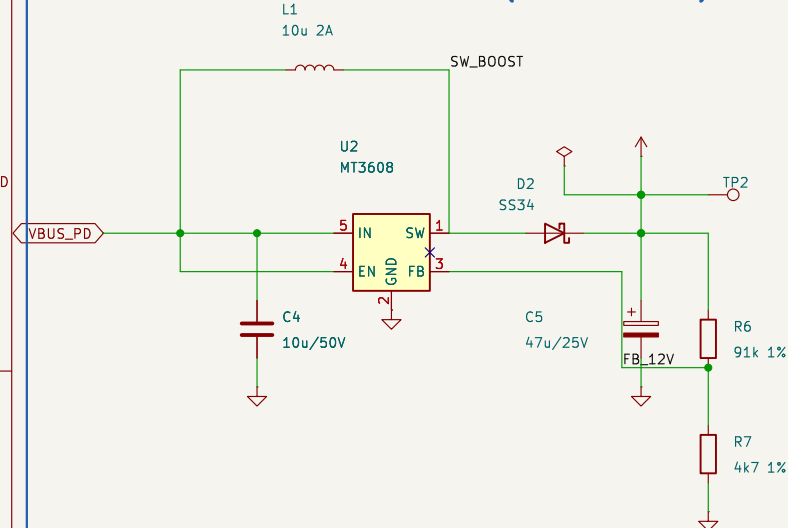
github.com/mohammadrezasafaeian (c) 2026 Mohammadreza Safaeian. All rights reserved.		
Designed and drawn by Mohammadreza Safaeian – m.re.safaeian@gmail.com		
Title page – specification, block diagram, sheet index		
Mohammadreza Safaeian – personal project		
Sheet: /		
File: usbc_pd_psu.kicad_sch		
Title: USB-C PD Bench PSU – 4-Switch Buck-Boost + PNP Linear Post-reg		
Size: A3	Date: 2026-04-25	Rev: v1.2
KiCad E.D.A. 9.0.2	Id: 1/3	

1 – USB-C INPUT, PROTECTION & PD TRIGGER



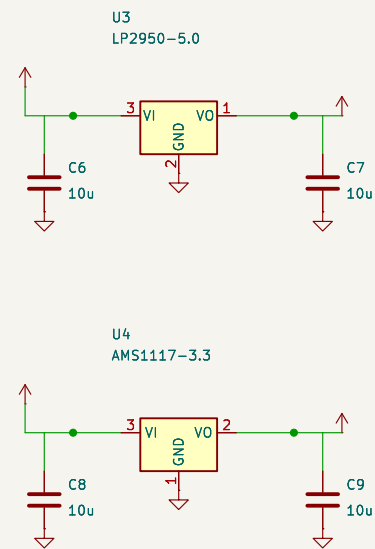
PD contract: CFG1/2/3 select 5 / 9 / 12 / 15 / 20 V. CFG1 is pulled high so a bare board requests 5 V before firmware runs. 500 ms after plug-in the MCU re-reads ADC_VBUS; no rise means a legacy 5 V port and the firmware clamps the output range instead of faulting. The PDO is re-selected as the setpoint changes (table on the title page), with the output disabled during renegotiation.

2 - +12 V GATE-DRIVE RAIL (MT3608 boost)



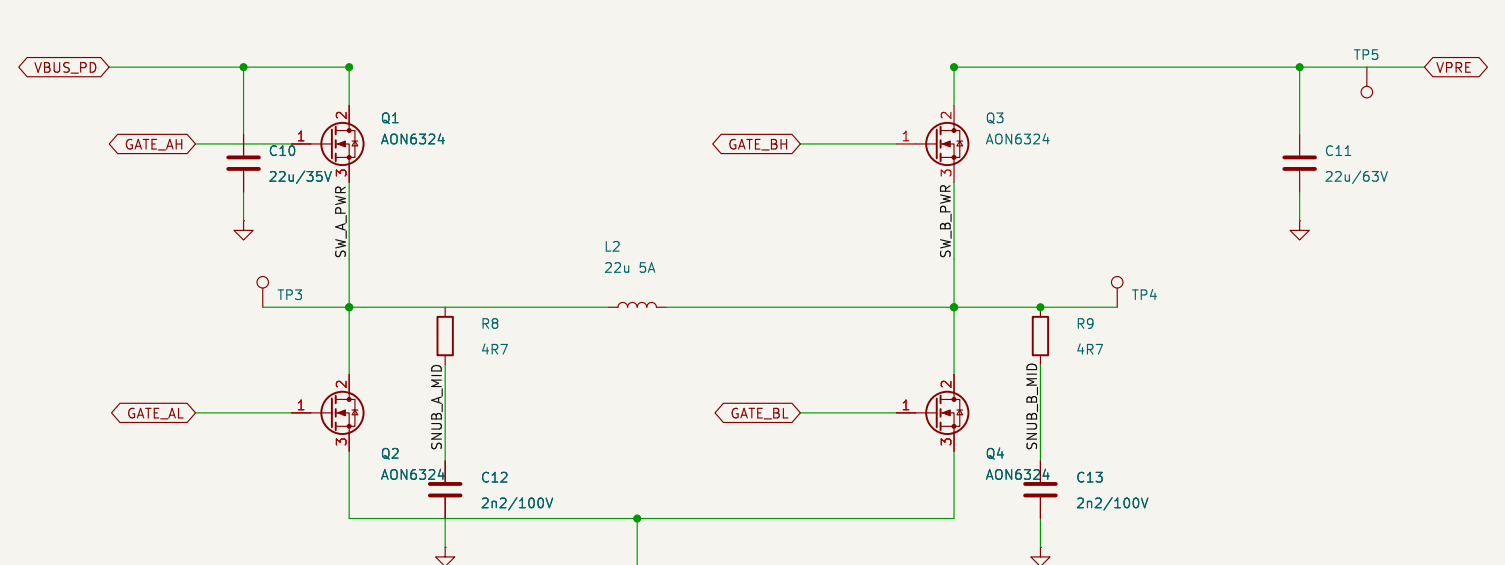
Why an on-board boost: the IRS2104 needs 10–12 V of gate drive (UVLO 8.9 V typ) to put the bridge FETs at their rated $R_{DS(on)}$. This 10 μ H / 5534 stage makes 12 V at 150 mA from any contract between 5 V and 20 V, so the drivers behave the same on a legacy 5 V port as on a 20 V one.

6 – LOGIC SUPPLIES



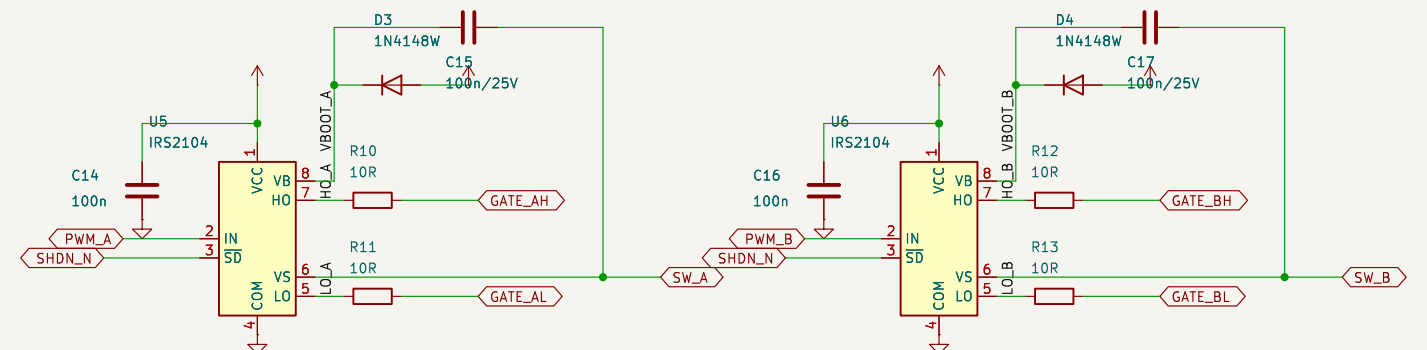
Fed from +12 V, not from VBUS: on a legacy port VBUS sits at ~4.75 V at the connector, below the LP2950 dropout for a 5 V output. Rails therefore come up identically on either port type.

3 – 4-SWITCH SYNCHRONOUS BUCK-BOOST POWER STAGE



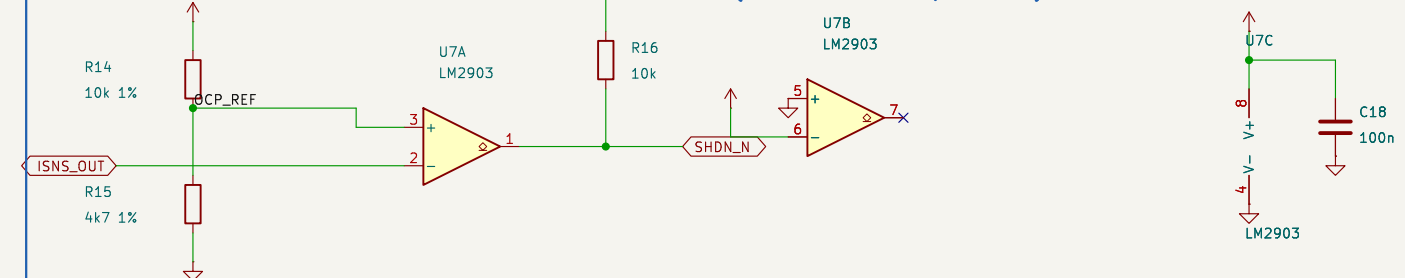
The 4R7 + 2n2 snubbers damp the switch-node ringing set by FET Coss and layout inductance. 60 V FETs, because a 20 V contract plus that ringing will exceed a 30 V part.

4 – HALF-BRIDGE GATE DRIVERS (IRS2104, 520 ns dead-time)



Each bootstrap capacitor returns to its own VS pin, which is tied to the switch node it references – that is what lets the cap refresh and the high-side FET turn on. 10R gate resistors slow the edges enough to keep the ringing in hand.

5 - HARDWARE OVER-CURRENT TRIP + LATCH (firmware-independent)



ISNS sits on the inverting input, so an over-current drives the open-collector output LOW and pulls SHDN_N down within ~1 us with no firmware involved. Both IRS2104 SD pins hang on SHDN_N, as does the MCU (open-drain), giving a wired-AND kill.

github.com/mohammadrezasafaeian | (c) 2026 Mohammadreza Safaeian. All rights reserved.

Designed and drawn by Mohammadreza Safaeian - m.re.safaeian@gmail.com

Sheet 1 of 2 - input, PD trigger, 12 V rail, buck-boost, drivers, OCP

Mohammadreza Safaeian – personal project

Sheet: /Power Path/

File: power.kicad_sch

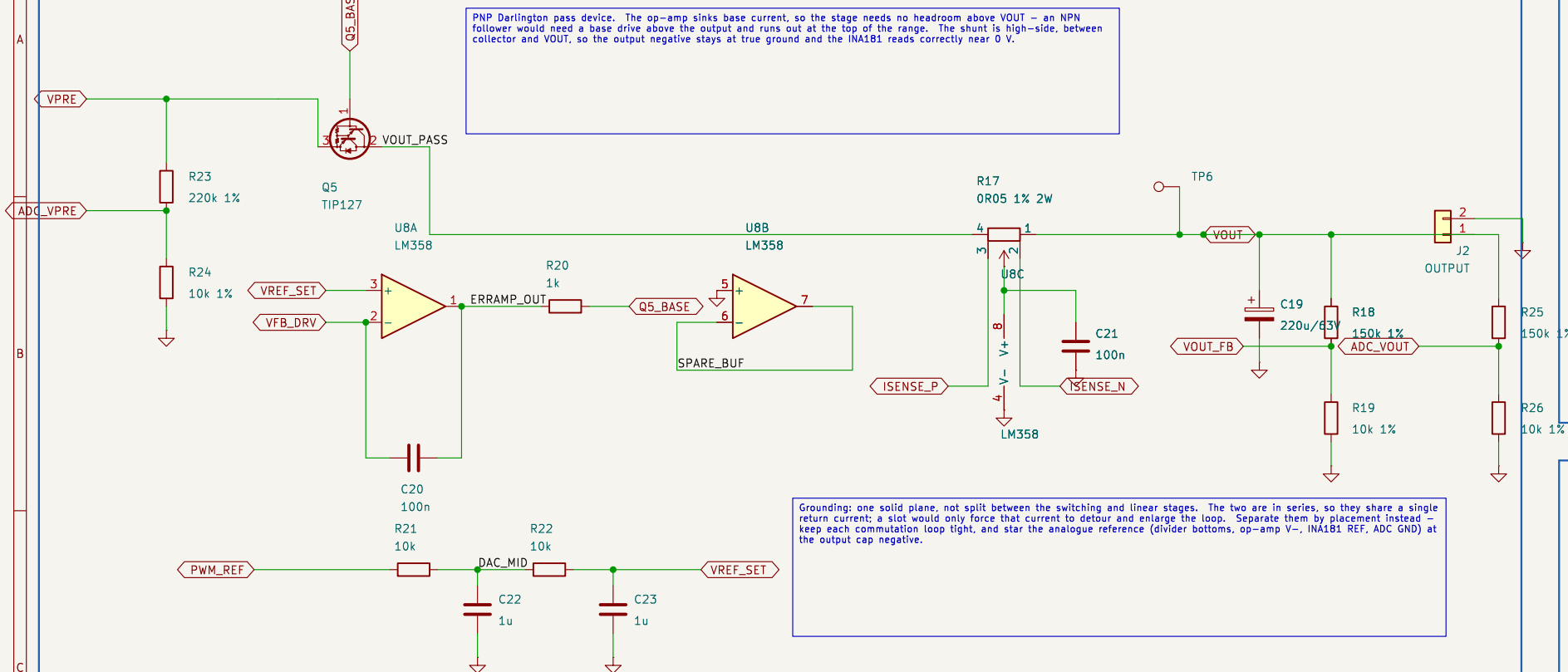
Title: USB-C PD Bench PSU - 4-Switch Buck-Boost + PNP Linear Post-

Size: A3	Date: 2026-04-25
----------	------------------

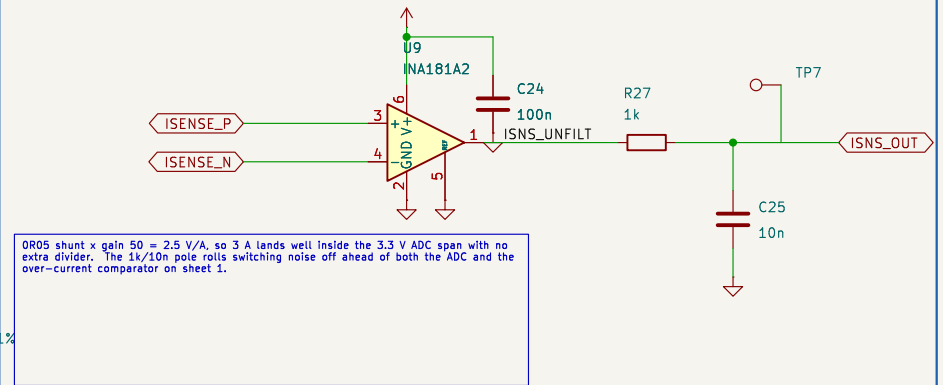
KiCad E.D.A. 9.0.2

RIC80 L.D.A. 9.0.2		10. 2/3
	7	8

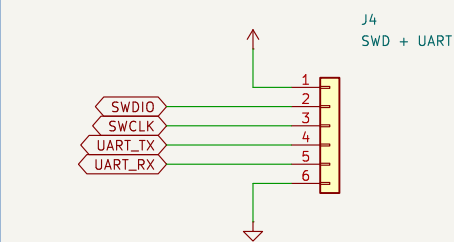
7 - PNP-DARLINGTON LINEAR POST-REGULATOR + OUTPUT



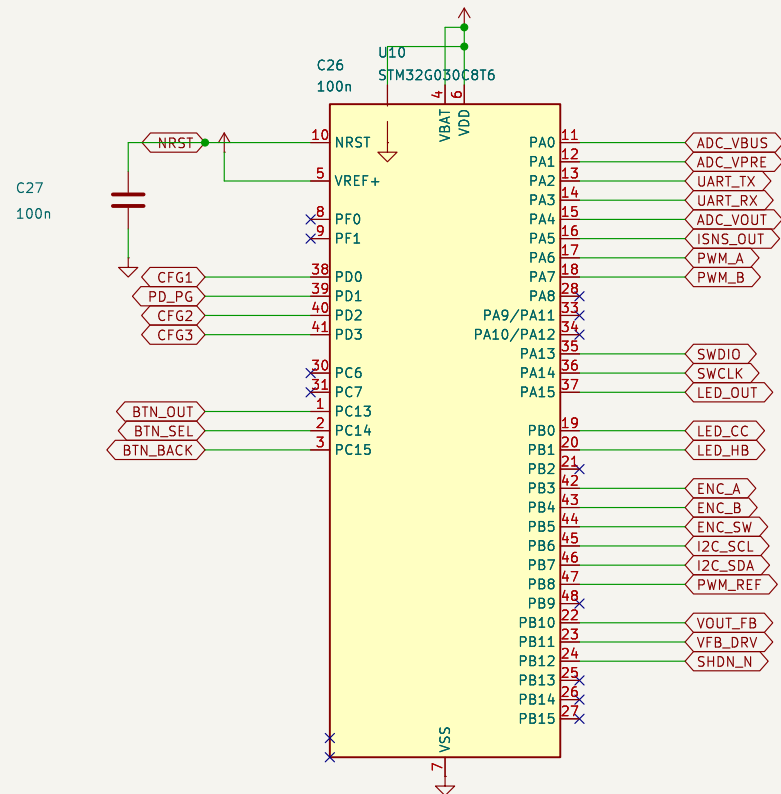
8 – CURRENT-SENSE AMPLIFIER (INA181A2, gain 50)



11 – SWD + UART HEADER



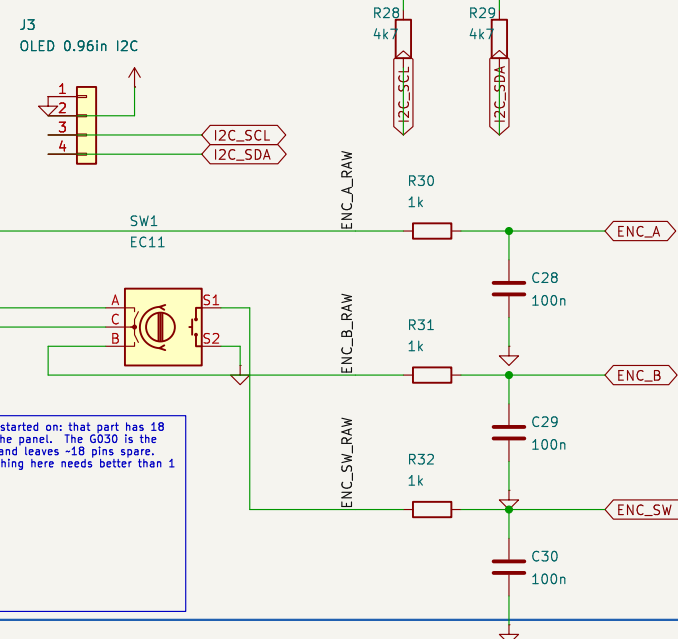
9 - MCU -- STM32G030C8T6 (LQFP-48)



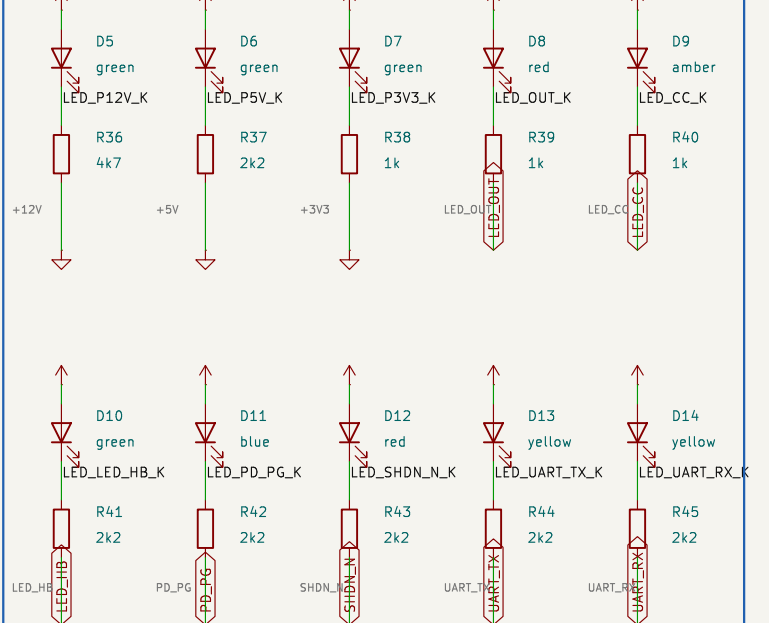
Indicator LEDs sink rather than source: +3V3 – LED – R – GPIO, so a pin left tri-stated at reset cannot back-feed the rail. PD_PG dark means the board fell back to a legacy 5 V port; SHDN_N lights on an over-current even with the MCU dead.

STM32G030C8T6, chosen over the CH32V003 the prototype started on: that part has 18 GPIO and 17 were already committed, leaving nothing for the panel. The G030 is the cheapest 48-pin Cortex-M0+ that is easy to source here and leaves -18 pins spare. Runs on the 16 MHz internal RC - no crystal, because nothing here needs better than 1% timing. SWD on PA13/PA14.

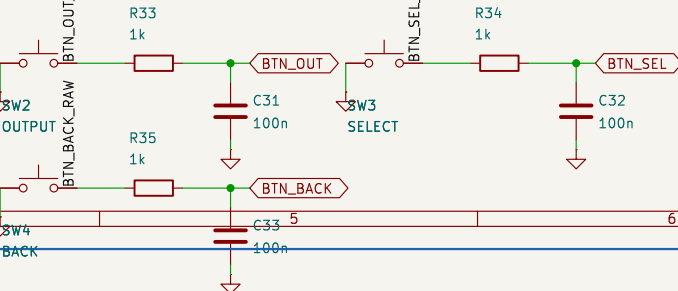
10 - UI: OLED + ROTARY ENCODER



12 – INDICATORS / LED TEST POINTS



13_RAW – FRONT PANEL



github.com/mohammadrezasafaeian | (c) 2026 Mohammadreza Safaeian. All rights reserved.

Designed and drawn by Mohammadreza Safaeian - m.re.safaeian@gmail.com

Sheet 2 of 2 – post-regulator, current sense, MCU, UI, programming

Mohammadreza Safaeian – personal project

Sheet: /Control and UI

File: control.kicad_sch
Title: USB-C PD Bench PSU - 4-Switch Buck-Boost + PNP Linear Post-

Size: A3

Date: 2026-04-25

Rev: v1.2

KiCad E.D.A. 9.0.2

Id: 3/3

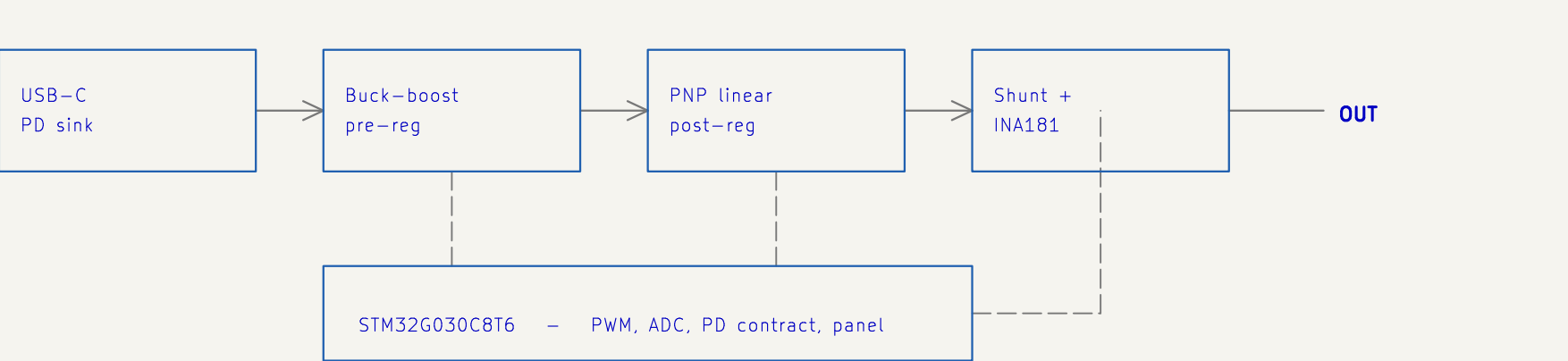
USB-C PD BENCH POWER SUPPLY

0 – 36 V / 0 – 3 A adjustable bench supply from a USB-C PD source

SPECIFICATION

Input	USB-C PD sink, 5 / 9 / 12 / 15 / 20 V contracts
Pre-regulator	4-switch synchronous buck-boost, 250 kHz
Post-regulator	PNP Darlington linear stage, tracking headroom
Output	0 – 36 V, 0 – 3 A
Set resolution	10 mV / 10 mA, PWM reference + 2-pole RC
Readback	voltage and current, 12-bit ADC
Protection	hardware over-current trip, latching
Modes	constant voltage, constant current, battery emulation
Control	STM32G030C8T6, OLED + rotary encoder + 3 keys

BLOCK DIAGRAM



REVISION HISTORY

Rev	MCU	Change
v0.3	CH32V003F4P6	first bench build; NPN Darlington follower
v0.4	CH32V003F4P6	pass device turned round to a PNP
v0.5	CH32V203C8T6	front panel added; the '003 runs out of pins
v0.6	CH32V203C8T6	hardware over-current trip
v1.1	STM32G030C8T6	redrawn as a hierarchy; on-sheet 12 V rail
v1.2	STM32G030C8T6	0–36 V range; analogue nets onto real ADC pins
v1.3	STM32G030C8T6	feedback closed, driver returns tied, sense rescaled

DESIGNED AND DRAWN BY

Mohammadreza Safaeian

Embedded Systems & Hardware Engineer

m.re.safaeian@gmail.com

github.com/mohammadrezasafaeian

(c) 2026 Mohammadreza Safaeian. All rights reserved.

Power Path

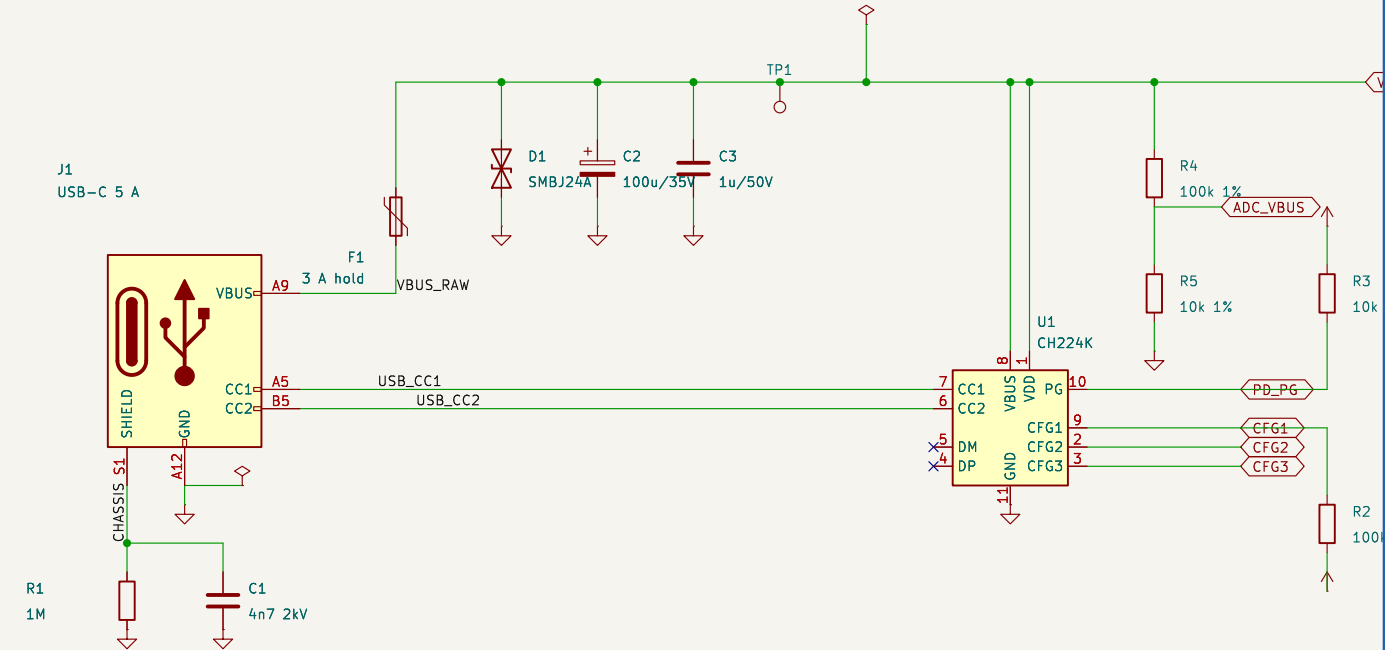
1	input / PD trigger
2	+12 V gate rail
3	buck-boost power stage
4	gate drivers
5	over-current trip

Control and UI

7	linear post-regulator
8	current sense
9	MCU
10	user interface
11	programming header

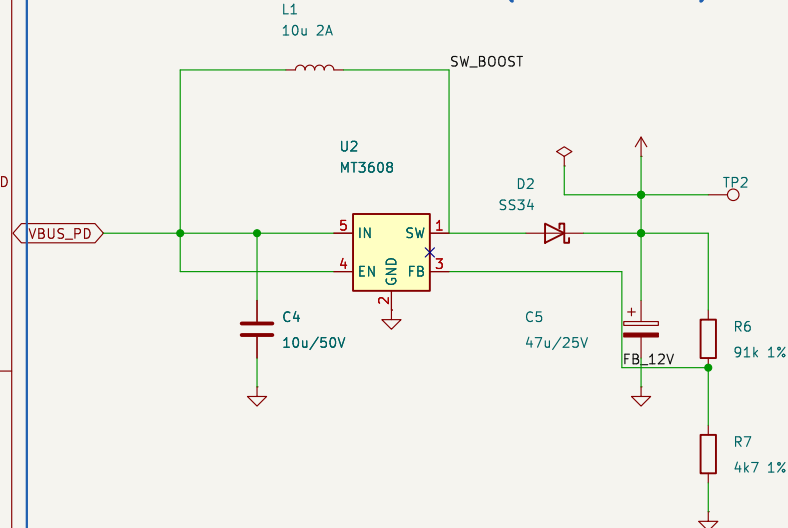
github.com/mohammadrezasafaeian (c) 2026 Mohammadreza Safaeian. All rights reserved.		
Designed and drawn by Mohammadreza Safaeian – m.re.safaeian@gmail.com		
Title page – specification, block diagram, sheet index		
Mohammadreza Safaeian – personal project		
Sheet: /		
File: usbc_pd_psu.kicad_sch		
Title: USB-C PD Bench PSU – 4-Switch Buck-Boost + PNP Linear Post-reg		
Size: A3	Date: 2026-05-25	Rev: v1.3
KiCad E.D.A. 9.0.2	Id: 1/3	

1 – USB-C INPUT, PROTECTION & PD TRIGGER



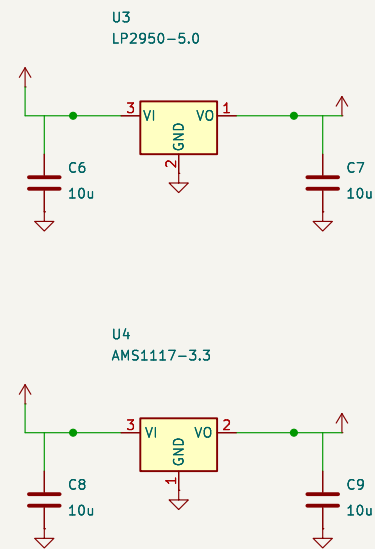
PD contract: CFG1/2/3 select 5 / 9 / 12 / 15 / 20 V. CFG1 is pulled high so a bare board requests 5 V before firmware runs. 500 ms after plug-in the MCU re-reads ADC_VBUS; no rise means a legacy 5 V port and the firmware clamps the output range instead of faulting. The PDO is re-selected as the setpoint changes (table on the title page), with the output disabled during renegotiation.

2 - +12 V GATE-DRIVE RAIL (MT3608 boost)



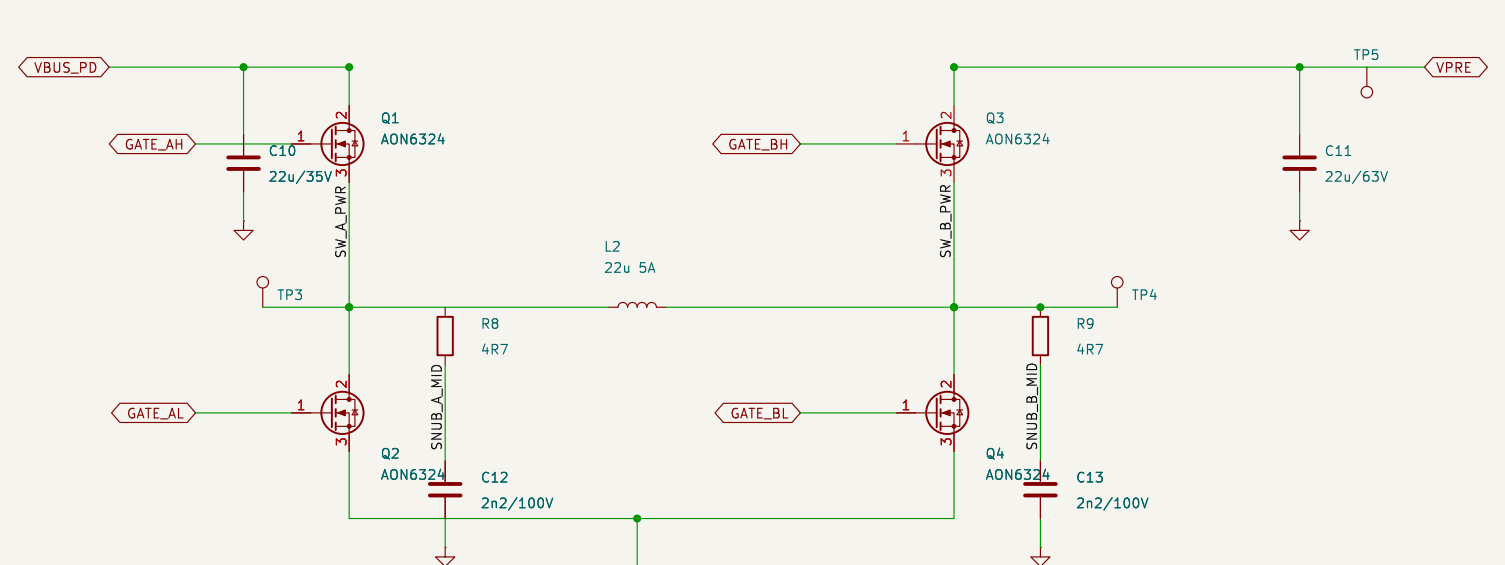
Why an on-board boost: the IRS2104 needs 10–12 V of gate drive (UVLO 8.9 V typ) to put the bridge FETs at their rated $R_{DS(on)}$. This 10 μ H / 5534 stage makes 12 V at 150 mA from any contract between 5 V and 20 V, so the drivers behave the same on a legacy 5 V port as on a 20 V one.

6 – LOGIC SUPPLIES



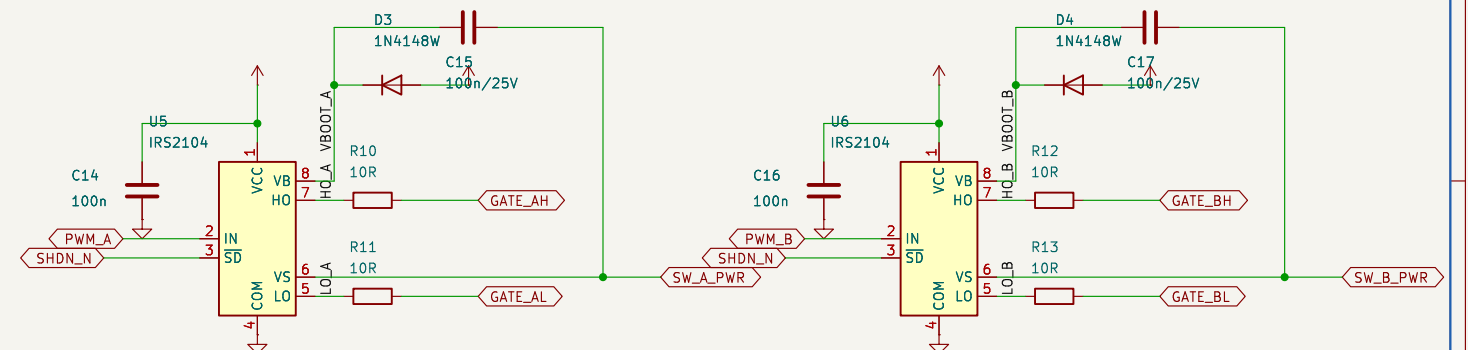
Fed from +12 V, not from VBUS: on a legacy port VBUS sits at ~4.75 V at the connector, below the LP2950 dropout for a 5 V output. Rails therefore come up identically on either port type.

3 – 4-SWITCH SYNCHRONOUS BUCK-BOOST POWER STAGE



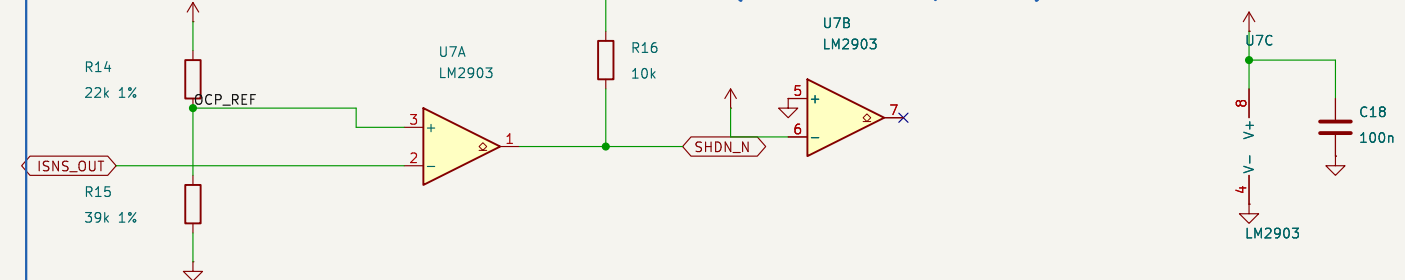
The 4R7 + 2n2 snubbers damp the switch-node ringing set by FET Coss and layout inductance. 60 V FETs, because a 20 V contract plus that ringing will exceed a 30 V part.

4 – HALF-BRIDGE GATE DRIVERS (IRS2104, 520 ns dead-time)



Each bootstrap capacitor returns to its own VS pin, which is tied to the switch node it references – that is what lets the cap refresh and the high-side FET turn on. 10R gate resistors slow the edges enough to keep the ringing in hand.

5 - HARDWARE OVER-CURRENT TRIP + LATCH (firmware-independent)



ISNS sits on the inverting input, so an over-current drives the open-collector output LOW and pulls SHDN_N down within ~1 us with no firmware involved. Both IRS2104 SD pins hang on SHDN_N, as does the MCU (open-drain), giving a wired-AND kill.

github.com/mohammadrezasafaeian | (c) 2026 Mohammadreza Safaeian. All rights reserved.

Designed and drawn by Mohammadreza Safaeian - m.re.safaeian@gmail.com

Sheet 1 of 2 - input, PD trigger, 12 V rail, buck-boost, drivers, OCP

Mohammadreza Safaeian – personal project

Sheet: /Power Path/

File: power.kicad_sch

Title: USB-C PD Bench PSU - 4-Switch Buck-Boost + PNP Linear Post-

Size: A3	Date: 2026-05-25
----------	------------------

KiCad E.D.A. 9.0.2

RIC80 L.D.A. 9.0.2		10. 2/3
	7	8

7 - PNP-DARLINGTON LINEAR POST-REGULATOR + OUTPUT

PNP Darlington pass device. The op-amp sinks base current, so the stage needs no headroom above VOUT – an NPN follower would need a base drive above the output and runs out at the top of the range. The shunt is high-side, between collector and VOUT, so the output negative stays at true ground and the INA1B1 reads correctly near 0 V.

Grounding: one solid plane, not split between the switching and linear stages. The two are in series, so they share a single return current; a slot would only force that current to detour and enlarge the loop. Separate them by placement instead – keep each commutation loop tight, and star the analogue reference (divider bottoms, op-amp V-, INA1B1 REF, ADC GND) at the output cap negative.

8 – CURRENT-SENSE AMPLIFIER (INA181A3, gain 100)

0R01 shunt $\times$ gain 100 = 1 V/A, so 3 A lands well inside the 3.3 V ADC span with no extra divider. The 1k/10n pole rolls switching noise off ahead of both the ADC and the over-current comparator on sheet 1.

11 - SWD + UART HEADER

Diagram illustrating the SWD + UART Header (J4) connections:

- Pin 1: SWDIO
- Pin 2: SWCLK
- Pin 3: UART_TX
- Pin 4: UART_RX
- Pin 5: Ground
- Pin 6: Ground

9 - MCU -- STM32G030C8T6 (LQFP-48)

Indicator LEDs sink rather than source: +3V3 - LED - GPIO, so a pin left tri-stated at reset cannot back-feed the rail. PD_PG dark means the board fed back to a legacy 5 V port; SHDN_N lights on an over-current even with the MCU dead.

STM32G030C8T6, chosen over the CH32V003 the price. GPIO0 and 17 were already committed, leaving nothing. cheapest 48-pin Cortex-M0+ that is easy to source. Runs on the 16 MHz internal RC - no crystal, but 1% timing. SWD on PA13/PA14.

10 - UI: OLED + ROTARY ENCODER

J3
OLED 0.96in I2C

SW1
EC11

A
C
B

S1
S2

ENC_A_RAW
ENC_B_RAW
ENC_SW_RAW

R29
4k7

R30
4k7

R31
1k

R32
1k

R33
1k

C30
100n

C31
100n

C32
100n

I2C_SCL

I2C_SDA

ENC_A

ENC_B

ENC_SW

Started on: that part has 18 pins. The G030 is the one I used and leaves 18 pins spare. Nothing here needs better than 1

12 – INDICATORS / LED TEST POINTS

The diagram illustrates the LED test points and their connections. The components are organized into five columns:

- Column 1:** LED D5 (green) at LED_P12V_K, resistor R37 (4k7) connected to +12V.
- Column 2:** LED D6 (green) at LED_P5V_K, resistor R38 (2k2) connected to +5V.
- Column 3:** LED D7 (green) at LED_P3V3_K, resistor R39 (1k) connected to +3V3.
- Column 4:** LED D8 (red) at LED_OUT_K, resistor R40 (1k) connected to LED_OV.
- Column 5:** LED D9 (amber) at LED_CC_K, resistor R41 (1k) connected to LED_CC.

Below these, another set of LEDs and resistors is shown:

- Column 6:** LED D10 (green) at LED_LED_HB_K, resistor R42 (2k2) connected to LED_HB.
- Column 7:** LED D11 (blue) at LED_PD_PG_K, resistor R43 (2k2) connected to PD_PG.
- Column 8:** LED D12 (red) at LED_SHDN_N_K, resistor R44 (2k2) connected to SHDN_N.
- Column 9:** LED D13 (yellow) at LED_UART_TX_K, resistor R45 (2k2) connected to UART_TX.
- Column 10:** LED D14 (yellow) at LED_UART_RX_K, resistor R46 (2k2) connected to UART_RX.

13 – FRONT PANEL

The diagram shows the front panel circuitry. It includes three buttons: **BTN_OUT**, **BTN_SEL**, and **BTN_BACK**. Each button is connected to a raw signal line (BTN_OUT_RAW, BTN_SEL_RAW, BTN_BACK_RAW) through a 1k resistor (R34, R35, R36). The raw signals are then connected to the buttons through 100nF capacitors (C33, C34, C35). The buttons are also connected to a common ground line labeled **BACK**. The **BACK** line is connected to a switch labeled **SW4** and a signal labeled **OUTPUT**. The **BTN_OUT** button is connected to a switch labeled **SW2** and a signal labeled **SELECT**. The **BTN_SEL** button is connected to a switch labeled **SW3**. The **BTN_BACK** button is connected to a switch labeled **SW4**.

Rev: v1.3
Id: 3/3